Deployment Guide
Pathlock Dynamic Access Controls - Data Scrambling

Version - 2026 Q1

Pathlock

info@pathlock.com | +1-908-782-5700 | 270 S. Main Street Flemington, NJ 08822
www.pathlock.com

Copyright Notice

Table of Contents

1 Overview

This guide details the deployment of Pathlock DAC for Data Scrambling in non-production systems. It specifies the necessary prerequisites and the sequential procedures for successful implementation, thereby ensuring data integrity and security while efficiently managing non-production environments.

1.1 Purpose

This guide describes the deployment of Pathlock DAC - Data Scrambling in non-production systems.

Important Information:

- Data scrambling is only permissible in non-production systems.
- Retaining a client backup before proceeding with data scrambling is advisable.
- Cluster tables are not supported.
- A field value, once scrambled, cannot be scrambled again.

1.2 Required Roles and Authorization Objects

The following PFCG Roles are essential for accessing the Pathlock DAC - Data Scrambling application:

PFCG Roles	Purpose	T-Code Assigned
`/APPSDM/DS_ADMIN`	This administrative role grants extensive access for configuring and executing data scrambling.	`/APPSDM/DS` – Data Scrambling configuration

The following authorization objects are required to access the Pathlock DAC - Data Scrambling UI:

Authorization Object	Activity	PFCG Roles
`Y&DS_CONF`	02 (Change) 03 (Display)	`/APPSDM/DS_ADMIN - Pathlock DAC`: Data Scrambling
`Y&DS_SCR`	04 (Prepare Staging) 05 (Process Scrambling)	`/APPSDM/DS_ADMIN - Pathlock DAC`: Data Scrambling

Note: Users require an additional PFCG role, which must be assigned in **T-Code `/N/APPSDM/DS_AUTHSETUP`**, to perform Data Scrambling.

2 Configuring Pathlock Data Scrambling

To perform data scrambling within a non-production system, you must configure Pathlock Data Scrambling using the Pathlock DAC: Data Scrambling tool. You can access this tool via transaction /N/APPSDM/DS or through the user menu path Pathlock DAC: Data Scrambling.

Pathlock Data Scrambling configurations are broadly categorized into two sections:

- Functional Configuration
- Technical Configuration

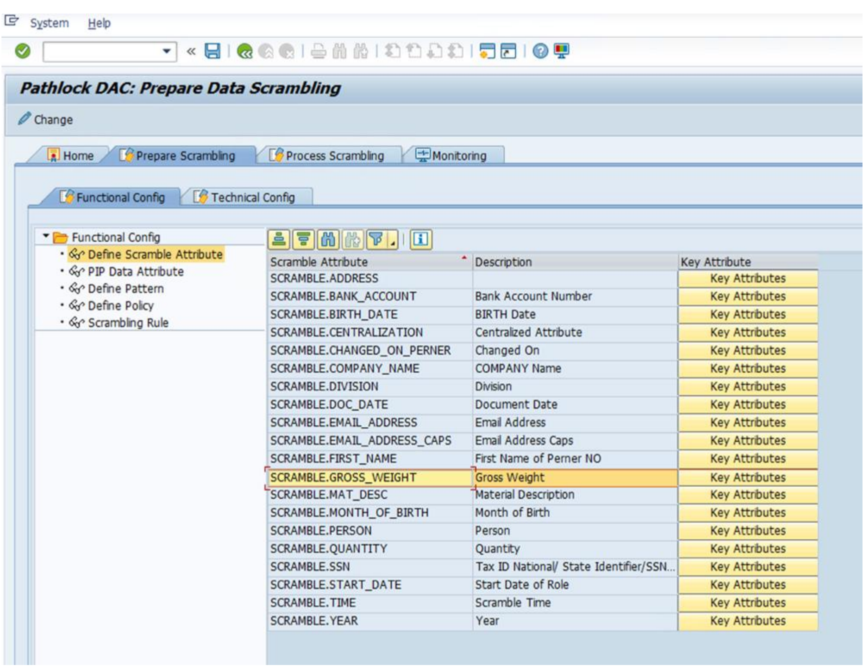
2.1 Functional Configuration

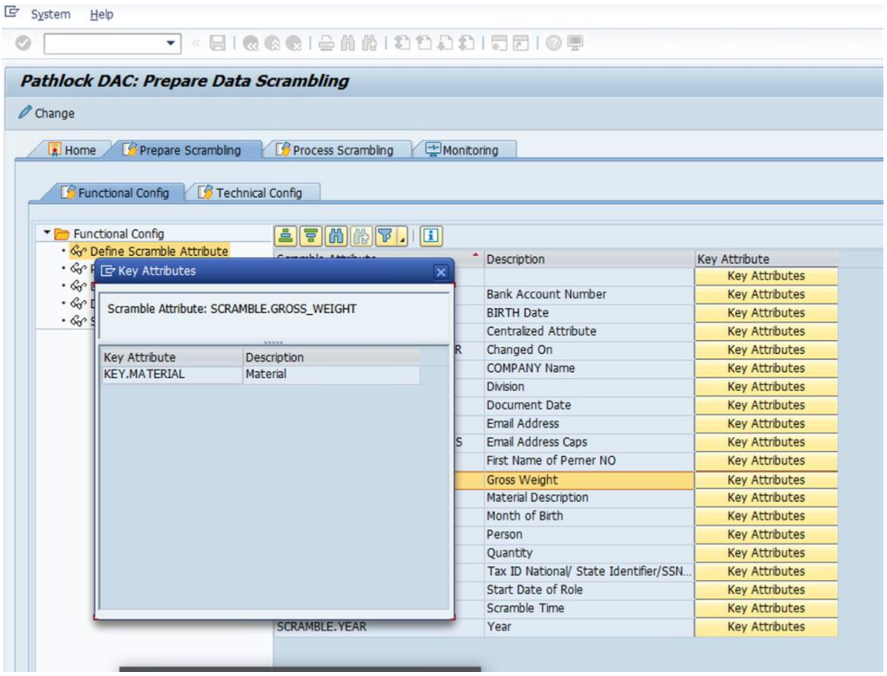
Business users perform functional configuration through the following steps:

- **Define Scramble Attribute:** This step defines the Scramble Attribute for data scrambling. Each Scramble Attribute requires the assignment of at least one Key Attribute.

  For example, to scramble the Gross Weight of a material, Gross Weight is defined as the Scramble Attribute, and Material Number is defined as the Key Attribute.

> **Note:** Use SCRAMBLE. as a prefix to define Scramble Attribute and KEY. as a prefix to define Key Attribute.

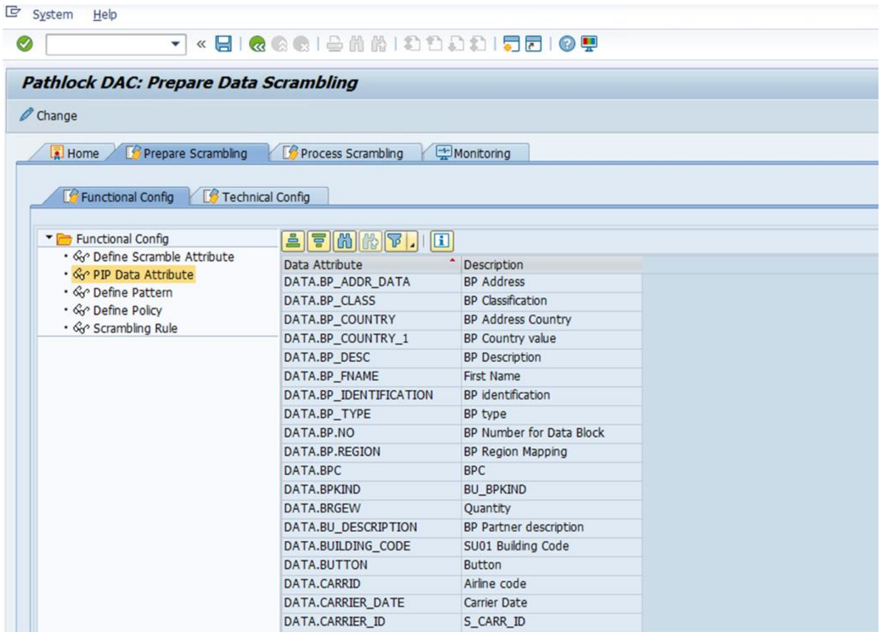




- **PIP Data Attribute:** This step defines data attributes for data objects such as Business Partner, Sales Document, Employee Salary, Country, etc. These attributes are required to define filter criteria. Additionally, DAC offers a customer-specific implementation that allows the derivation of Data attributes as required.

  For example, a Data Attribute such as Material Group can be defined and used in filter criteria to scramble the Gross Weight for materials belonging to specific material groups.

Note: Use DATA. as a prefix to define all Data Attributes.

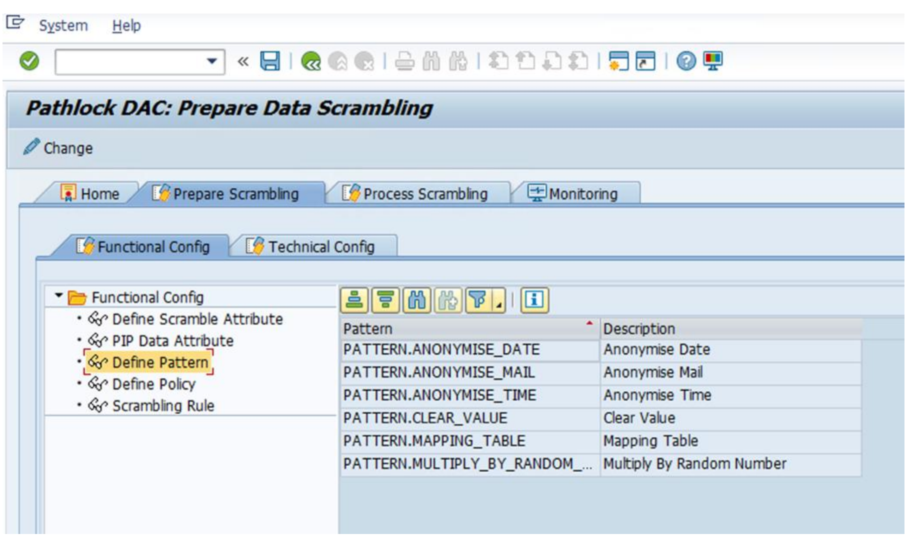


- **Define Pattern:** This step defines a pattern to determine how the final output appears after scrambling. The pattern guides the transformation of the original data into its scrambled form.

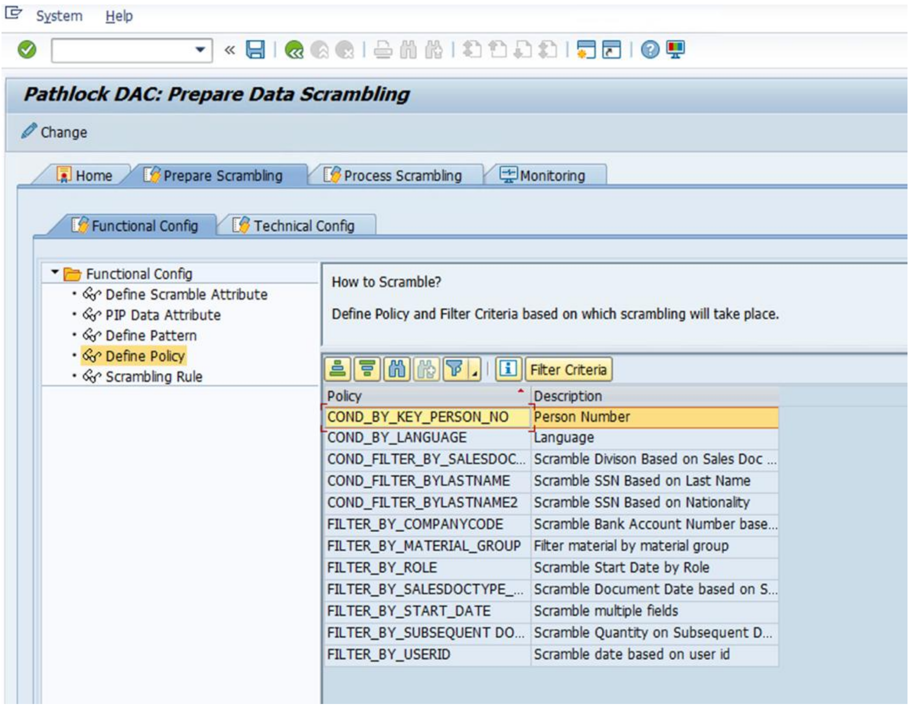
  The solution delivers the following patterns:

  - `PATTERN.ANONYMISE_DATE`
  - `PATTERN_MAPPING_TABLE`
  - `PATTERN_MULTIPLY_BY_RANDOM_NUMBER`

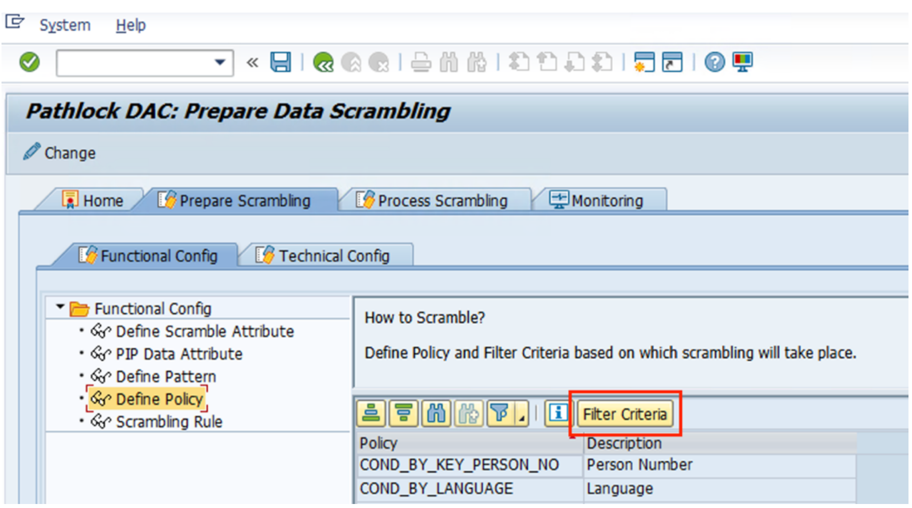
> **Note:** Use PATTERN. as a prefix to define a pattern.

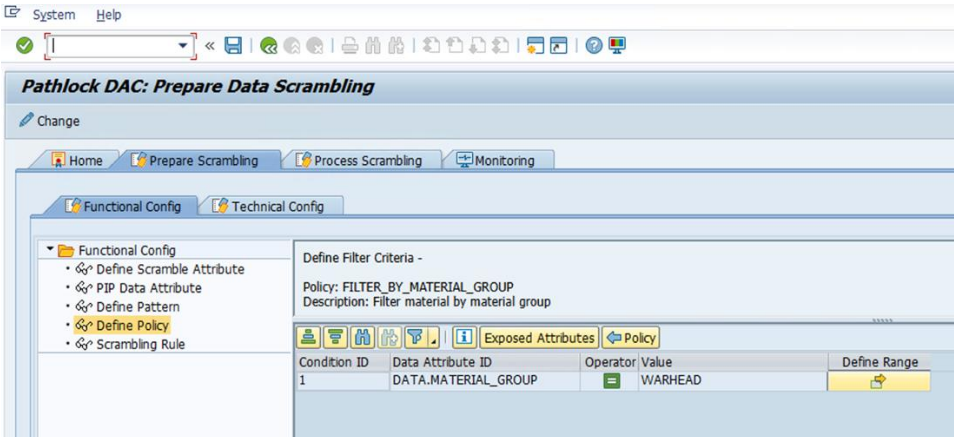


- **Define Policy:** The system dynamically evaluates these policies based on the available **Data Attributes** during runtime.



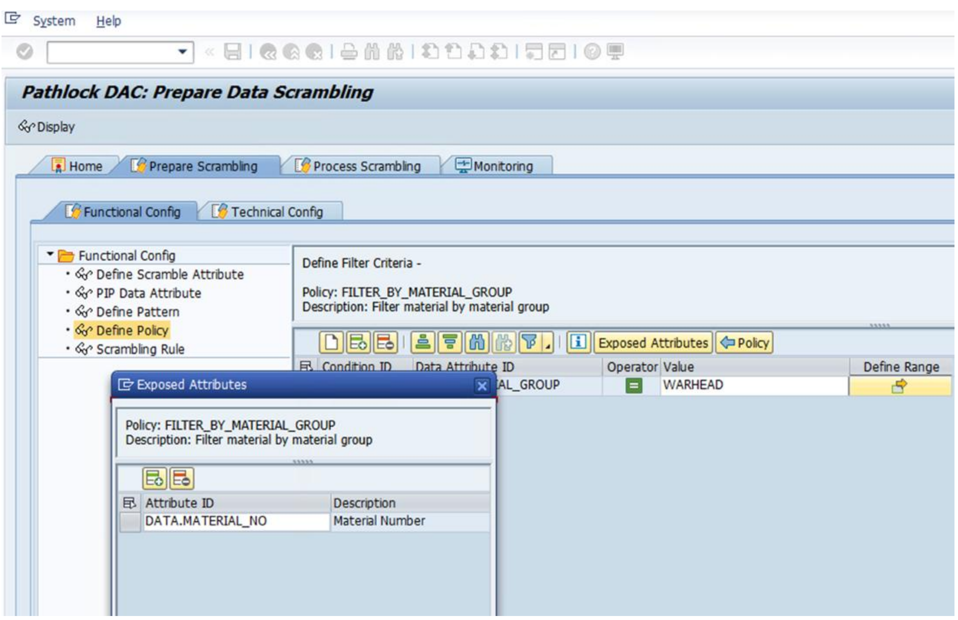
- **Define Filter Criteria:** This step involves setting up rules based on one or more conditions that utilize Data Attributes within the system. The system dynamically evaluates the filter criteria at runtime.



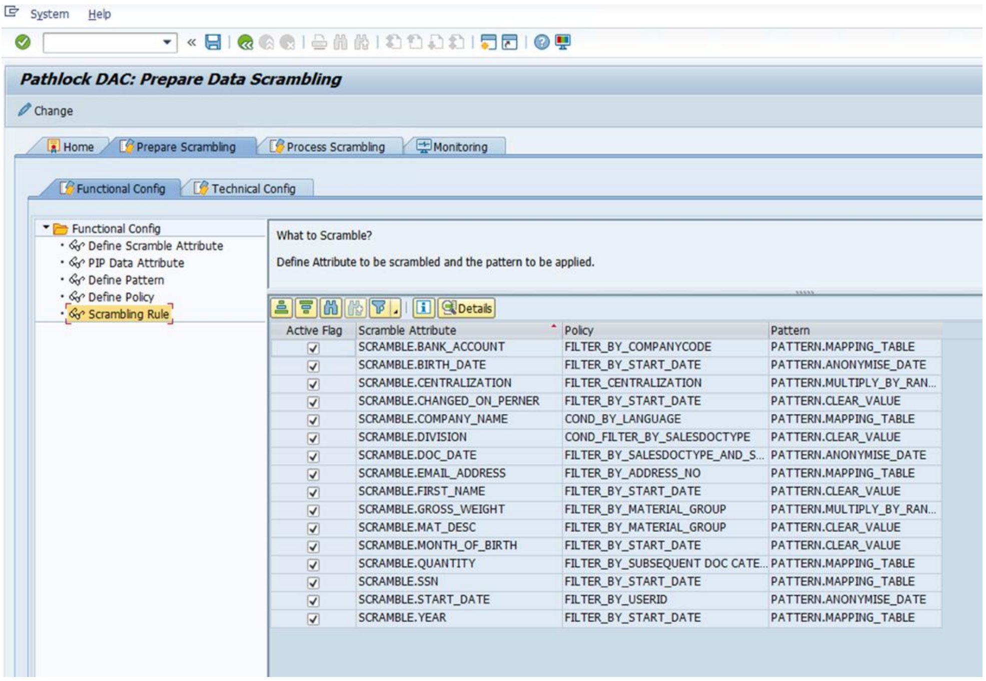


- **Exposed Attributes:** This step allows linking additional attributes to derive the actual Data Attribute used in filter criteria through the Data Attribute Badi# (/APPSDM/DATA_ATTRIBUTES).

  For example, when scrambling the Material Description based on Material Type in MAKT table since 'Material Type' is not available in MAKT table, you can use the additional attribute 'Material Number' (which is available in the table) to derive 'Material Type'. This can be achieved by implementing the Data Attribute Badi#(/APPSDM/DATA_ATTRIBUTES).



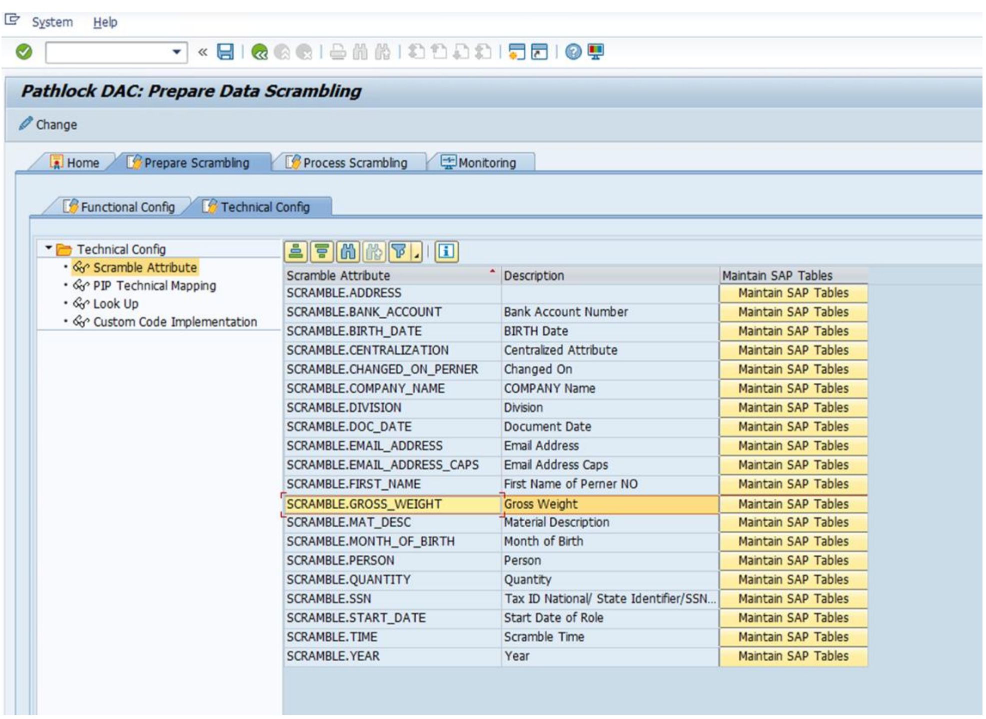
- **Scrambling Rule:** This step assigns a Policy and Pattern to the Scramble Attribute to define the execution of scrambling actions.



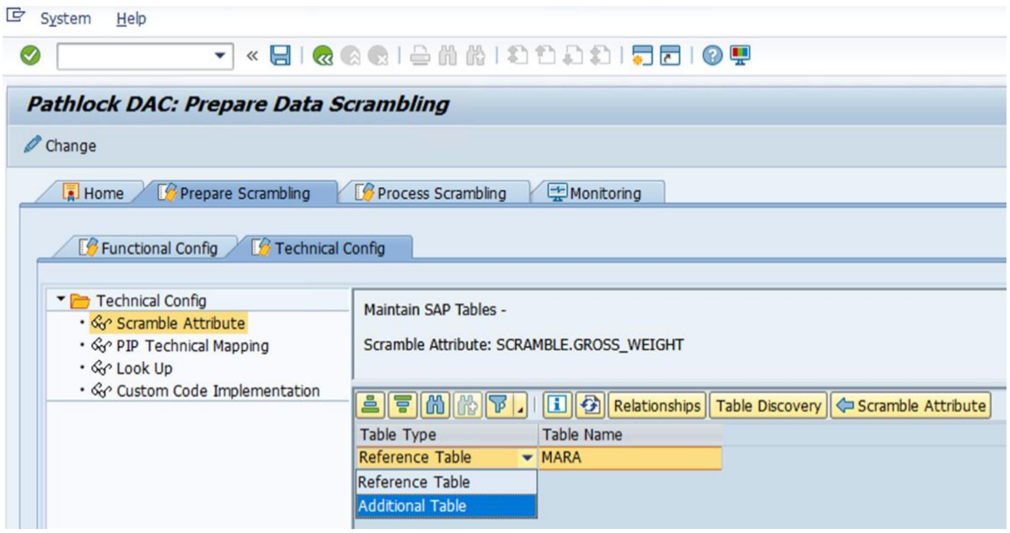
2.2 Technical Configuration

Technical users perform the technical configuration through the following steps:

- **Scramble Attribute:** This step maintains SAP tables for each Scramble Attribute.

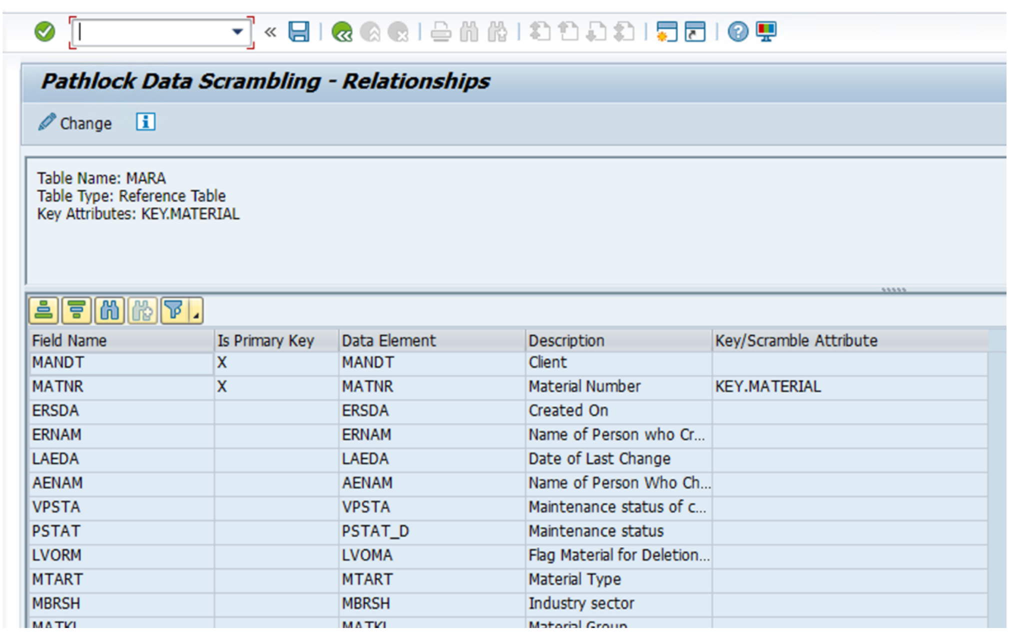


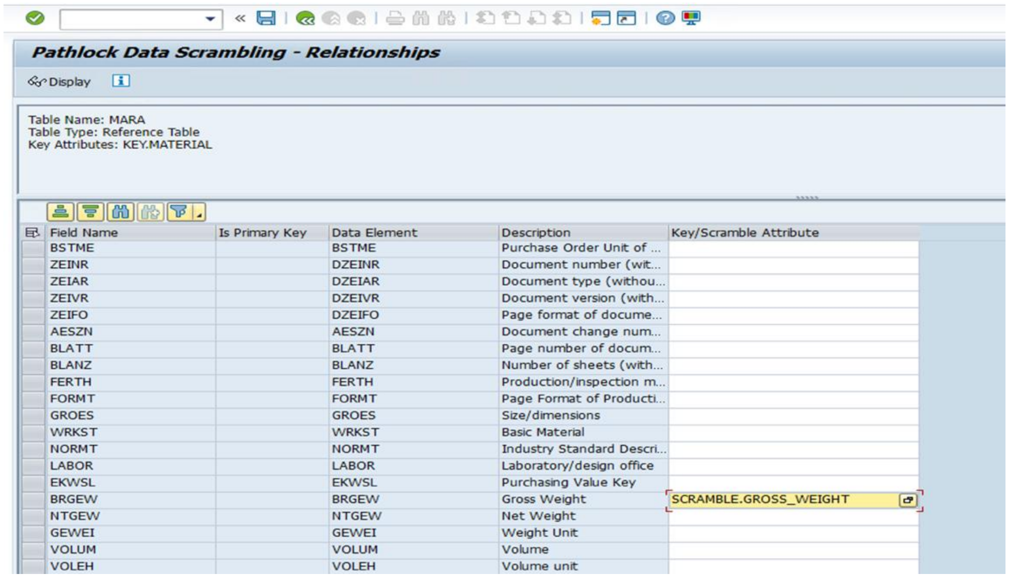
- **Maintain SAP Tables:** This step maintains the reference table for the Scramble Attribute, which the system uses to prepare Staging Data. You can also maintain additional tables if required.



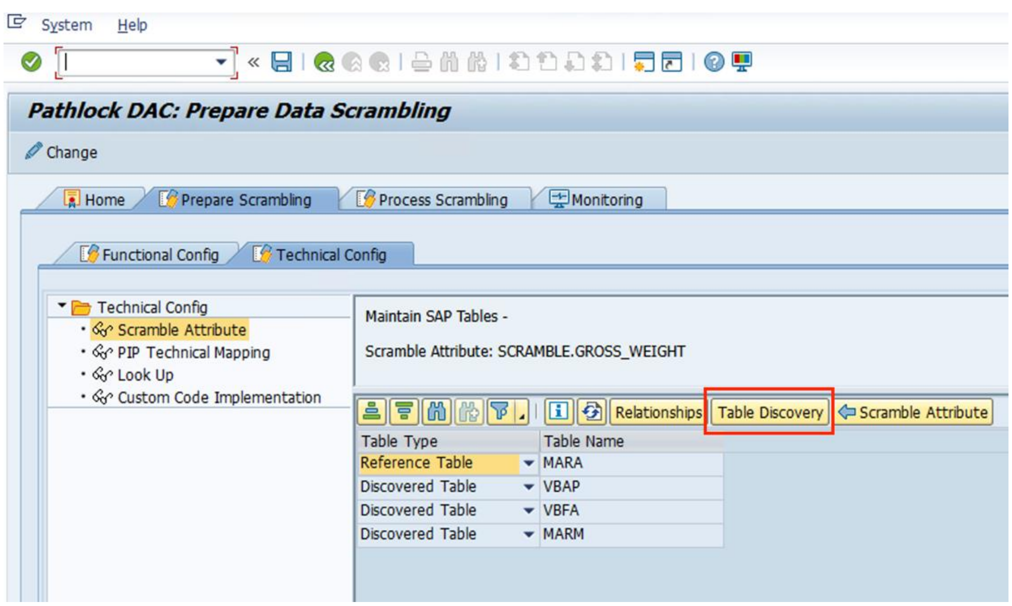
- **Maintain Relationships:** After setting up the reference table, link the reference table fields to the Scramble Attribute and the Key Attributes.

  For example, Material Number is linked to the Key Attribute, and Gross Weight is linked to the Scramble Attribute.

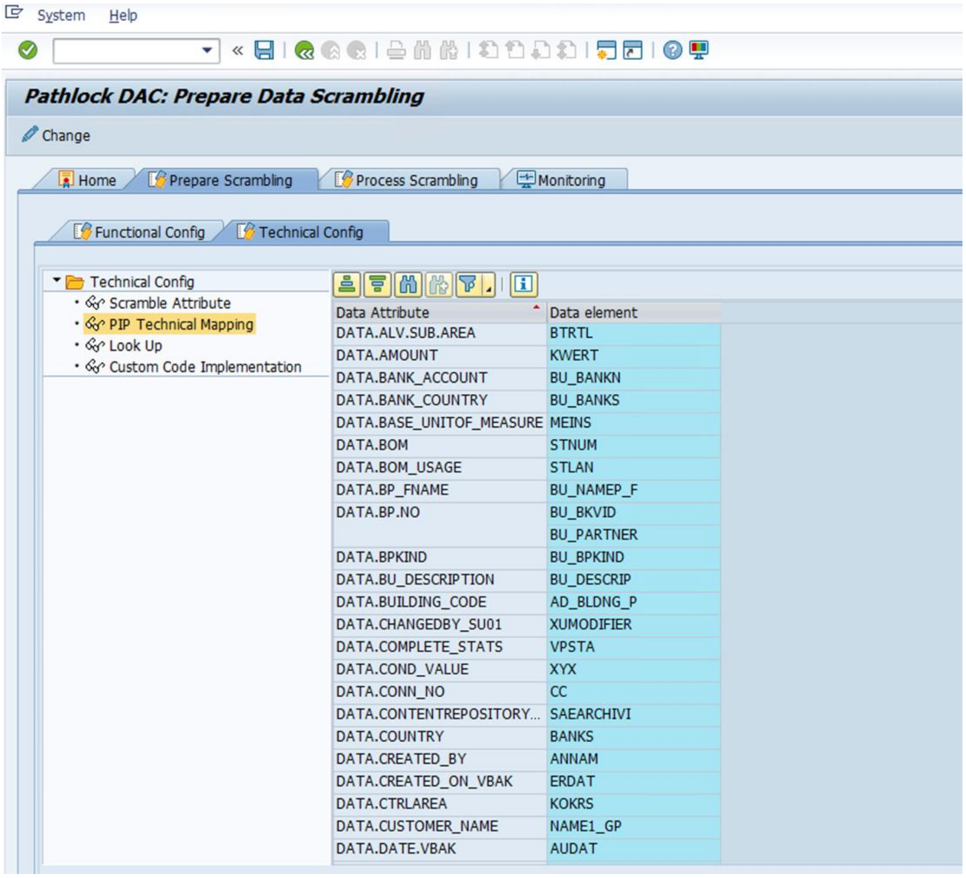




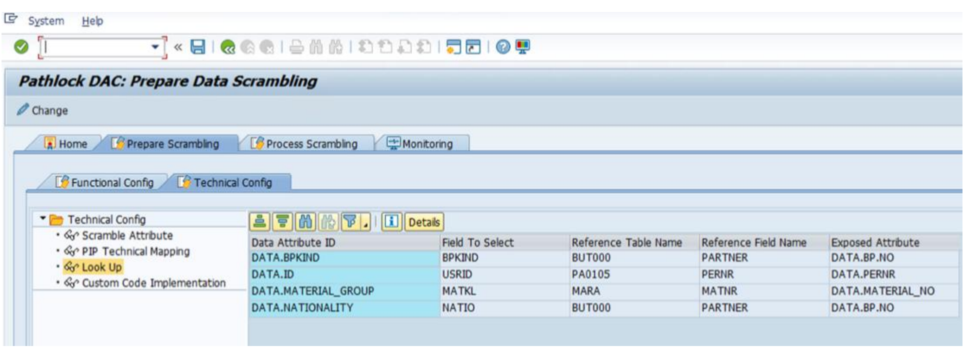
- **Table Discovery:** After linking the Scramble Attribute and Key Attribute, perform Table Discovery to generate related tables where fields assigned to the Key Attribute and Scramble Attribute coexist. If the system does not discover any tables, you can add them manually as additional tables. The system will consider the reference table, discovered tables, and additional tables during the scrambling process.



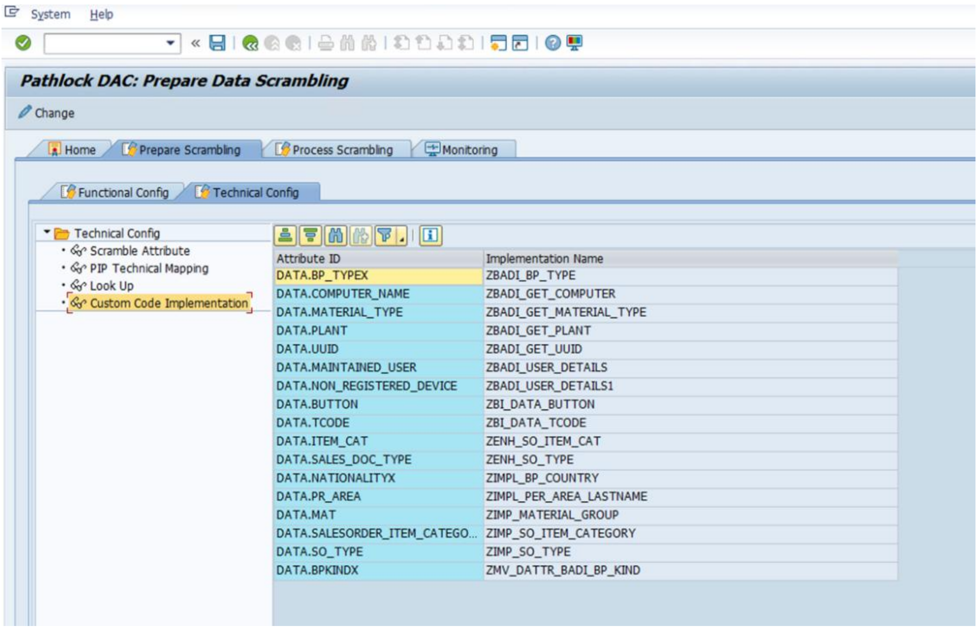
- **PIP Technical Mapping:** This step assigns Data Elements to the Data Attributes using this configuration node.



- **Look Up:** This functionality enables the configuration of SELECT queries directly within the Look Up configuration, eliminating the need for a BADI.

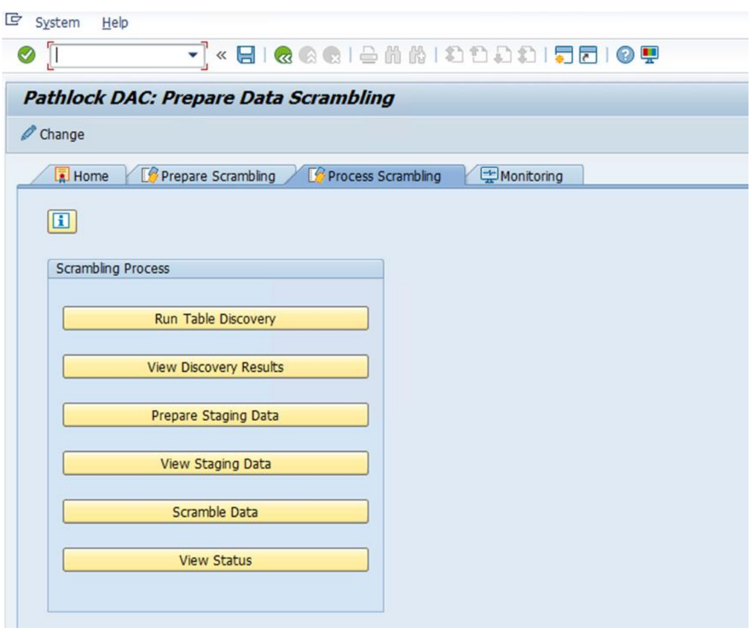


- **Custom Code Implementation:** This step displays Data Attribute BADI implementations that are applied through custom code. Implement Data Attribute BADI# /APPSDM/DATA_ATTRIBUTES to derive Data Attribute values.

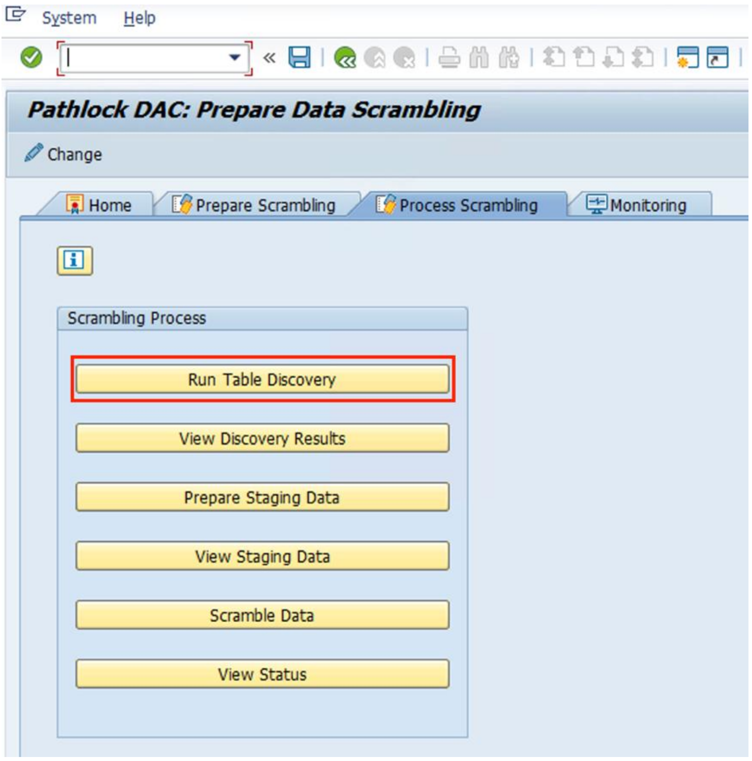


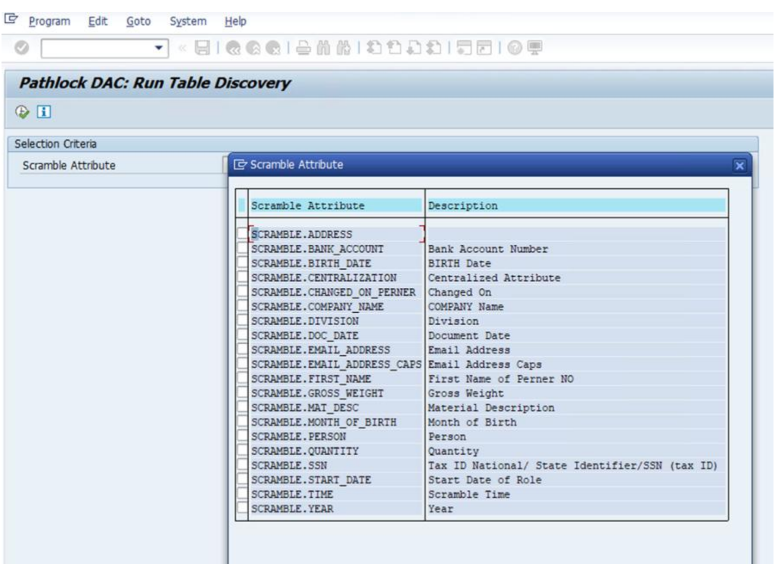
2.3 Process Data Scrambling

To proceed with Staging and Scrambling the attributes, navigate to the Process Data Scrambling tab.

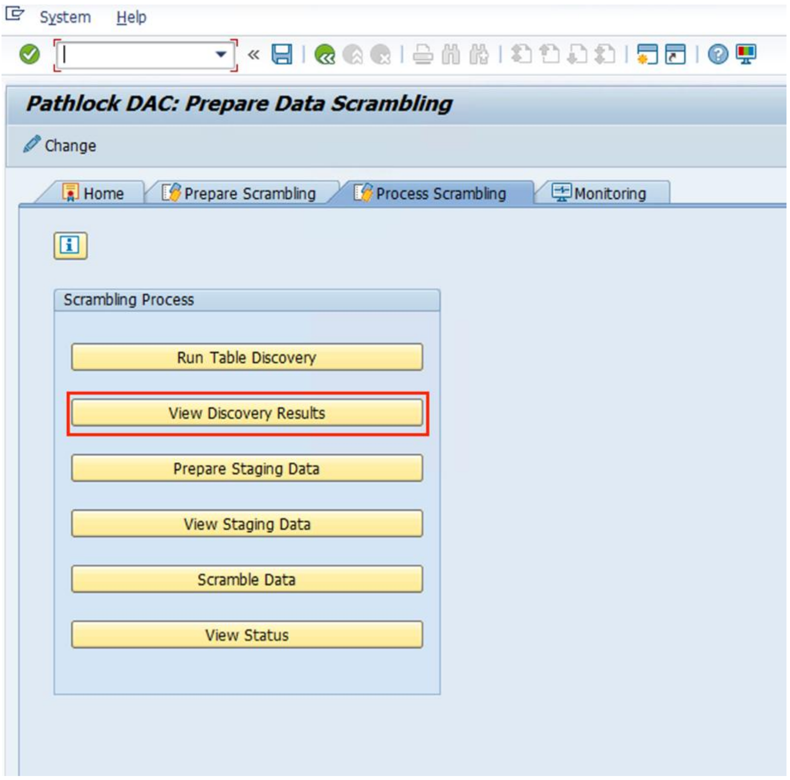


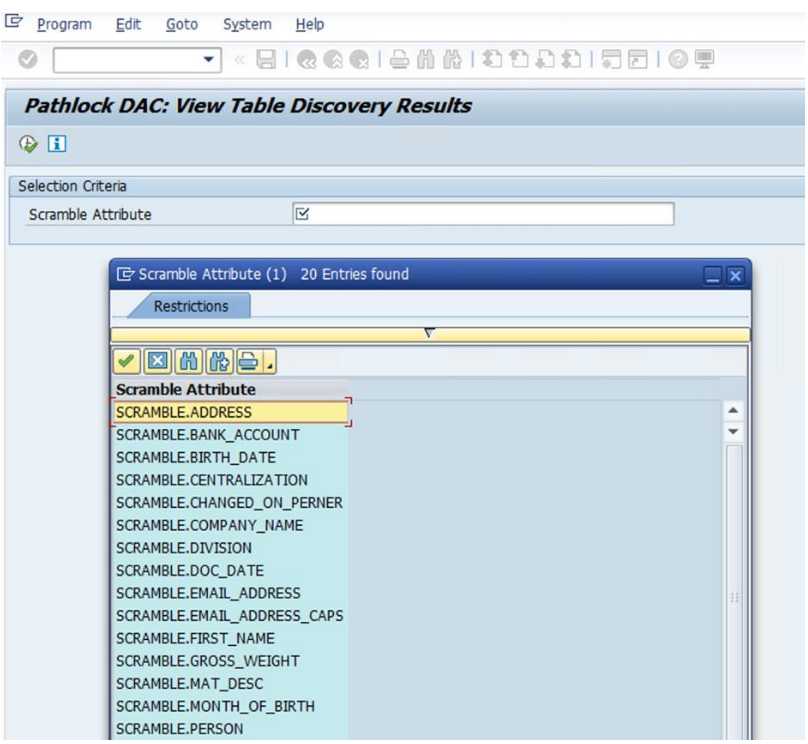
- **Run Table Discovery:** Generates related tables where the Key Attribute and Scramble Attribute coexist.



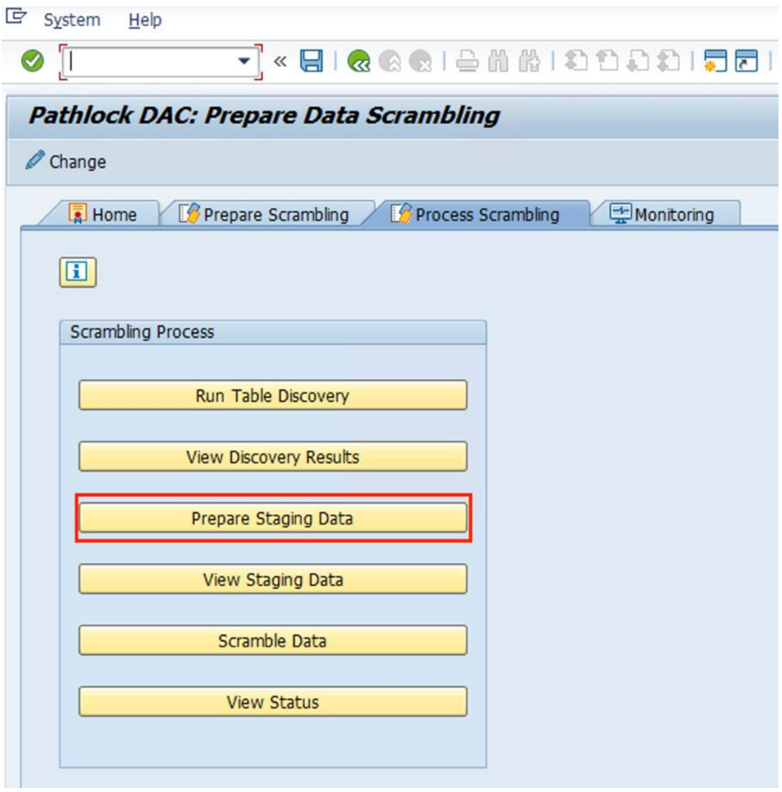


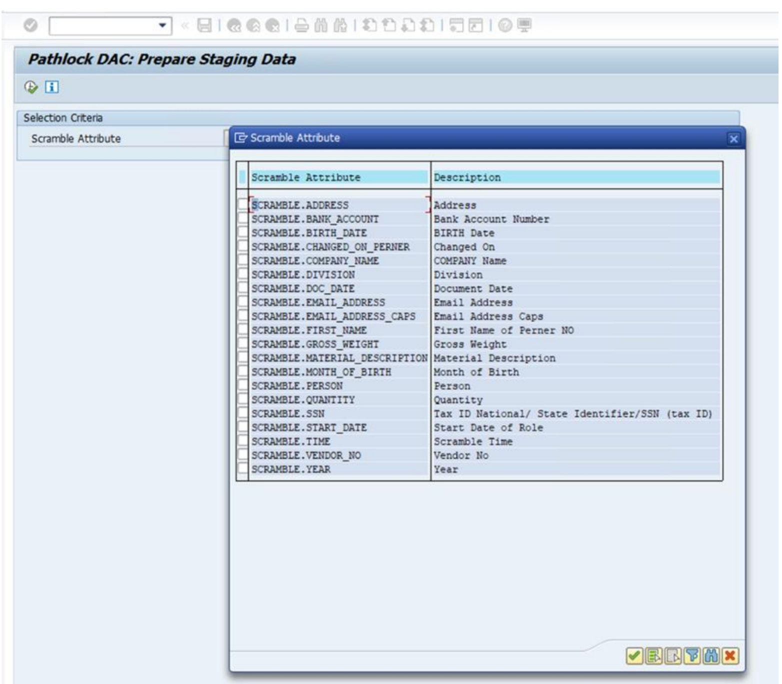
- **View Discovery Results:** Displays the reference, discovered, and additional tables associated with the specified Scramble Attribute, if maintained.


System Help
Pathlock DAC: Prepare Data Scrambling
Change
Home
Prepare Scrambling
Process Scrambling
Monitoring
Scrambling Process
Run Table Discovery
View Discovery Results
Prepare Staging Data
View Staging Data
Scramble Data
View Status

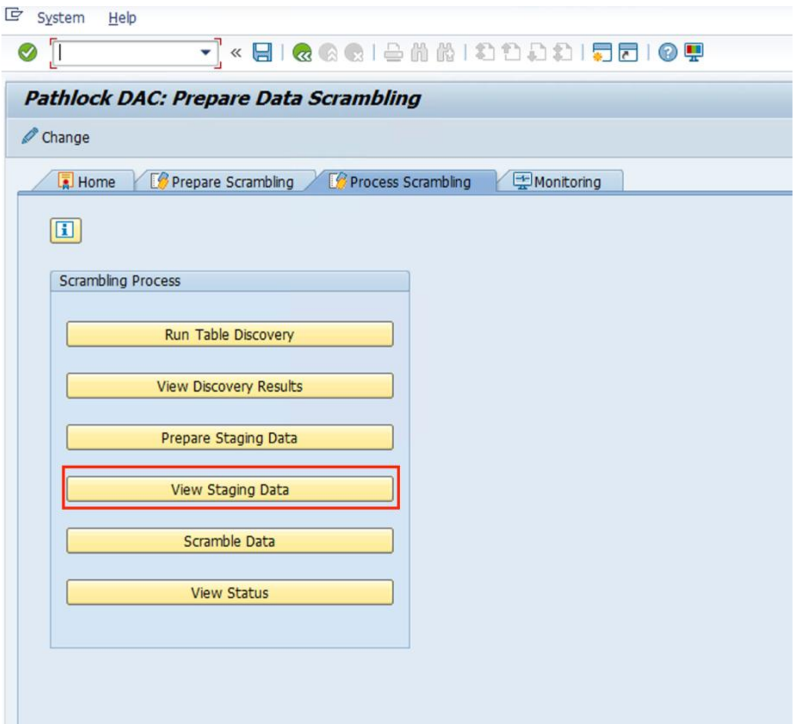

Program Edit Goto System Help
Pathlock DAC: View Table Discovery Results
Selection Criteria
Scramble Attribute
Scramble Attribute (1) 20 Entries found
Restrictions
Scramble Attribute
SCRAMBLE.ADDRESS
SCRAMBLE.BANK_ACCOUNT
SCRAMBLE.BIRTH_DATE
SCRAMBLE.CENTRALIZATION
SCRAMBLE.CHANGED_ON_PERNER
SCRAMBLE.COMPANY_NAME
SCRAMBLE.DIVISION
SCRAMBLE.DOC_DATE
SCRAMBLE.EMAIL_ADDRESS
SCRAMBLE.EMAIL_ADDRESS_CAPS
SCRAMBLE.FIRST_NAME
SCRAMBLE.GROSS_WEIGHT
SCRAMBLE.MAT_DESC
SCRAMBLE.MONTH_OF_BIRTH
SCRAMBLE.PERSON

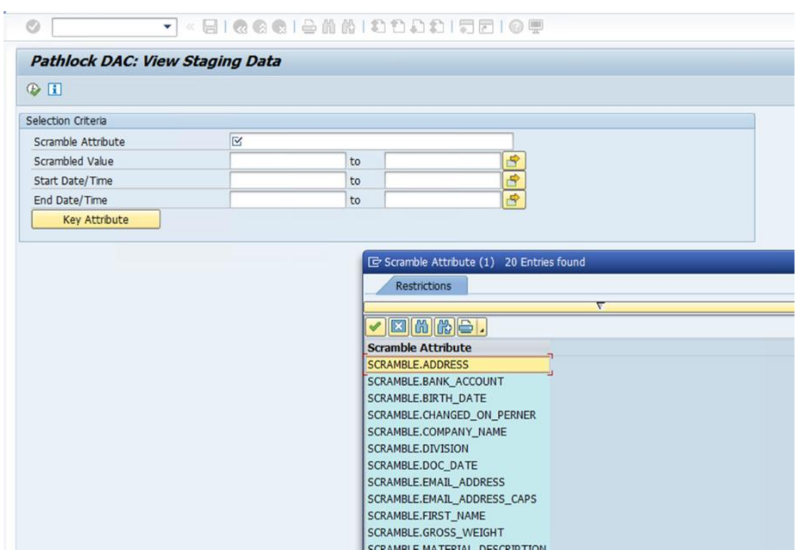
- **Prepare Staging Data:** Select the Scramble Attribute to stage the data. The job will be scheduled, and data will be sent to Staging area.



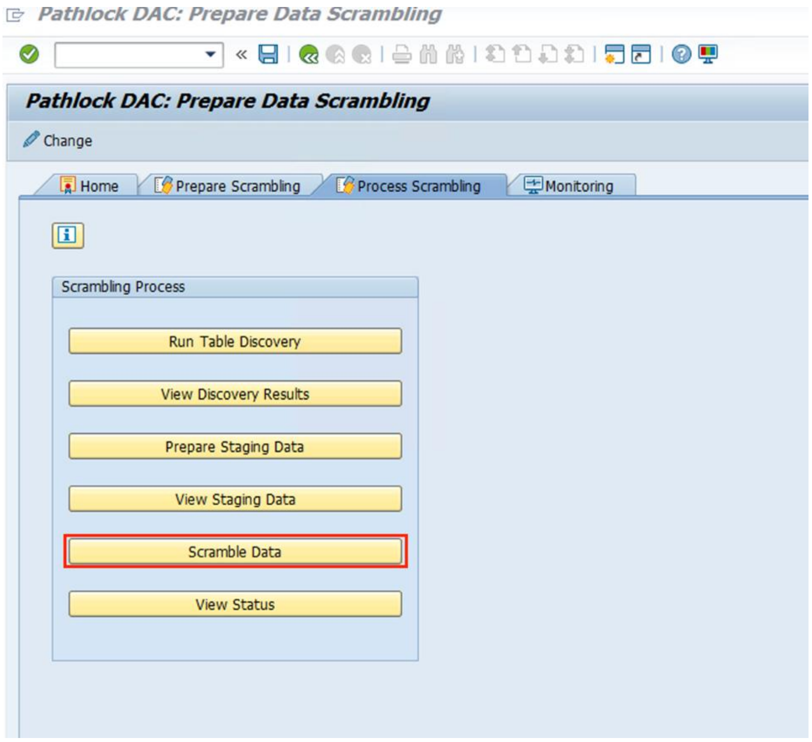


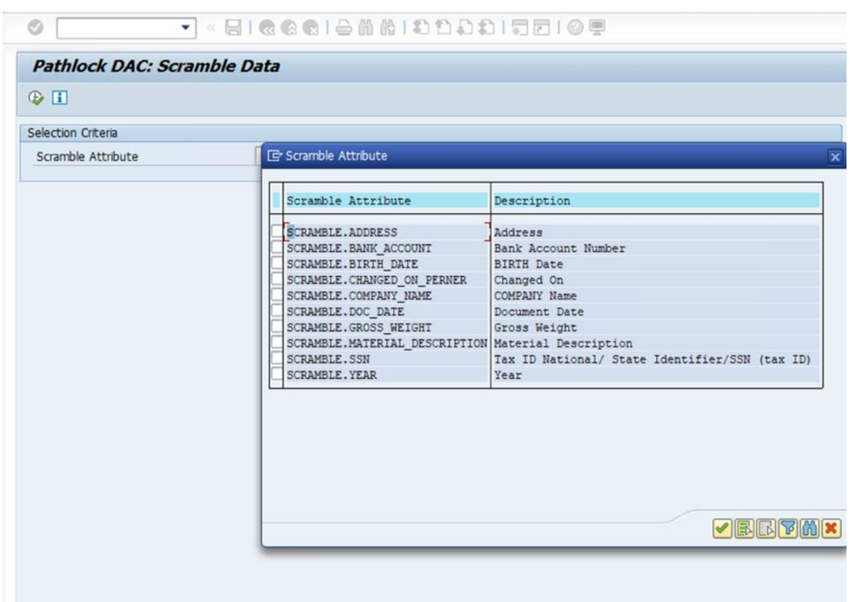
- **View Staging Data:** Select the Scramble Attribute to review the staged data before scrambling each Key Attribute.



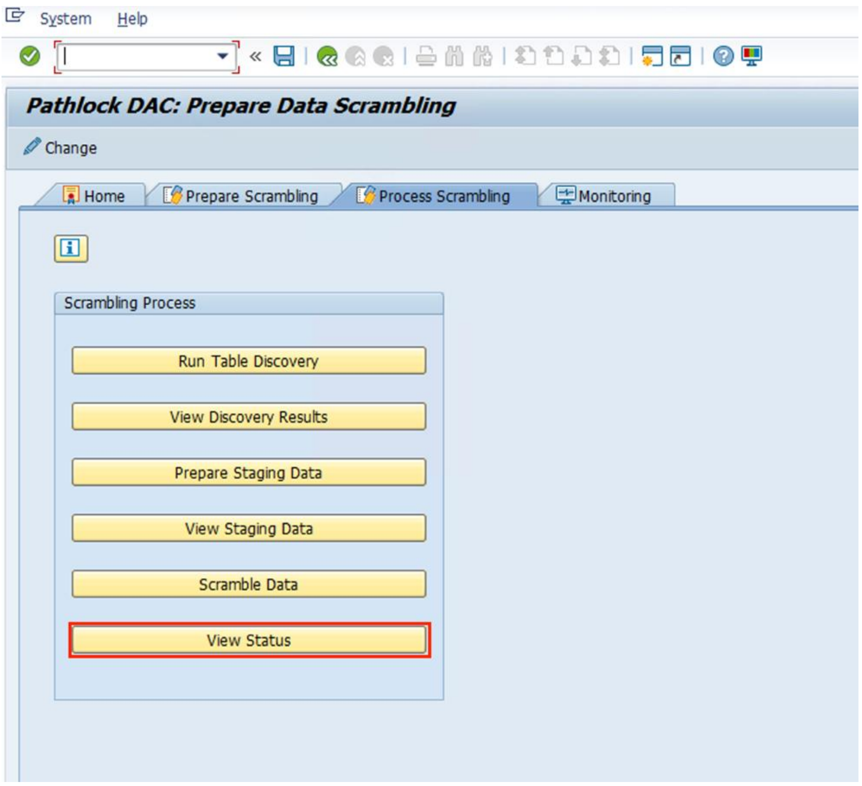


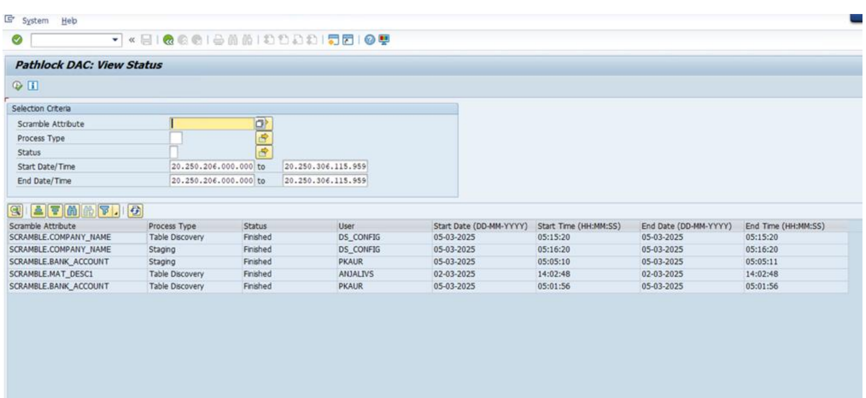
- **Scramble Data:** Select the Scramble Attribute to perform Data Scrambling.





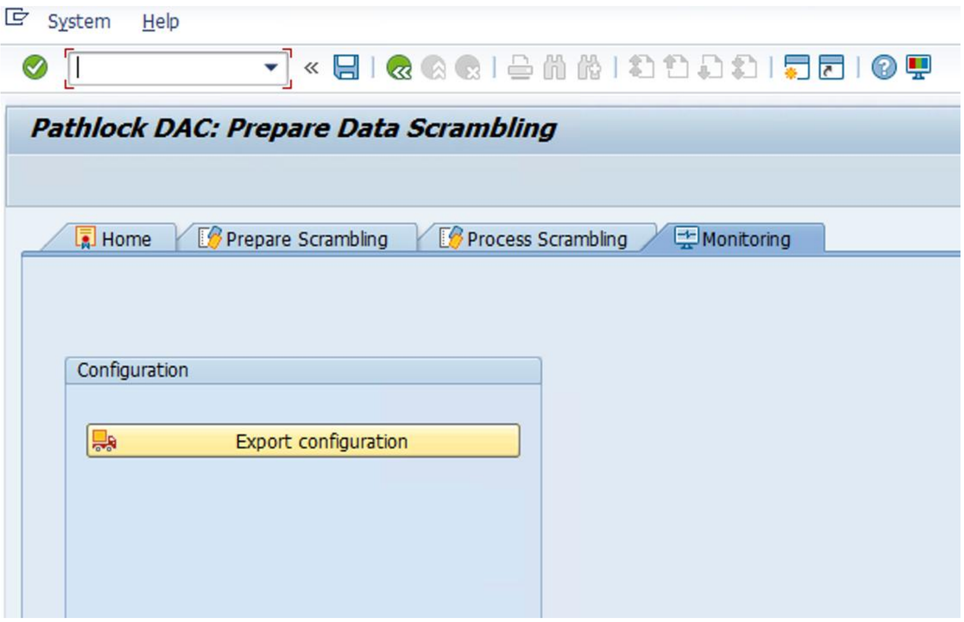
- **View Status:** To view Staging, Scrambling and Table Discovery job status, select the Scramble Attribute and select one of the available options.





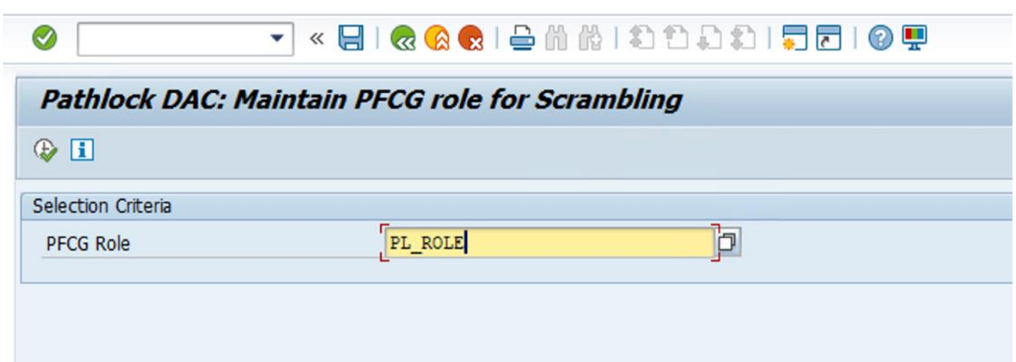
2.4 Monitoring

To export the Data Scrambling configuration via a Transport Request, navigate to the Monitoring tab and click the **Export configuration** button.



2.5 Maintain PFCG role for Scrambling

To maintain PFCG role for Scrambling, go to T-Code - /APPSDM/DS_AUTHSETUP and maintain the existing or new PFCG role, and the same role must be assigned to user to authorize for “Scramble Data”.



Note: This is a one-time setting. Users require this role to execute Data scrambling. PL_ROLE is used as an example; separate role must be maintained.

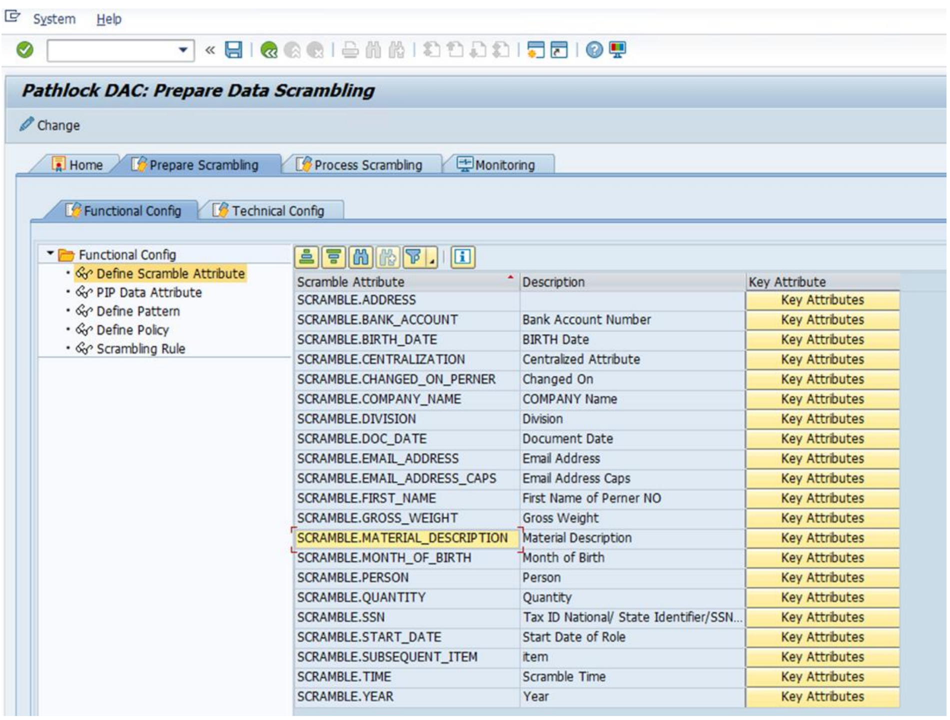
3 Sample Implementation

This section describes the use cases of the implementation.

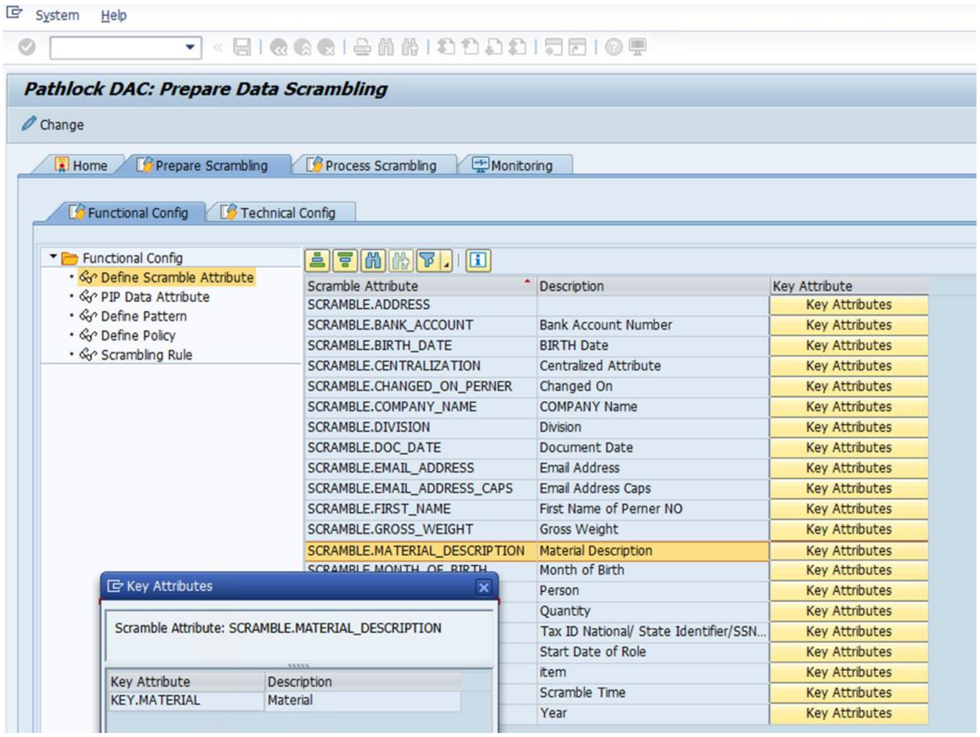
3.1 Use Case 1: Scrambling Material Description Based on Material Group and Mapping Table Pattern

To scramble 'Material Description', the following configuration is required:

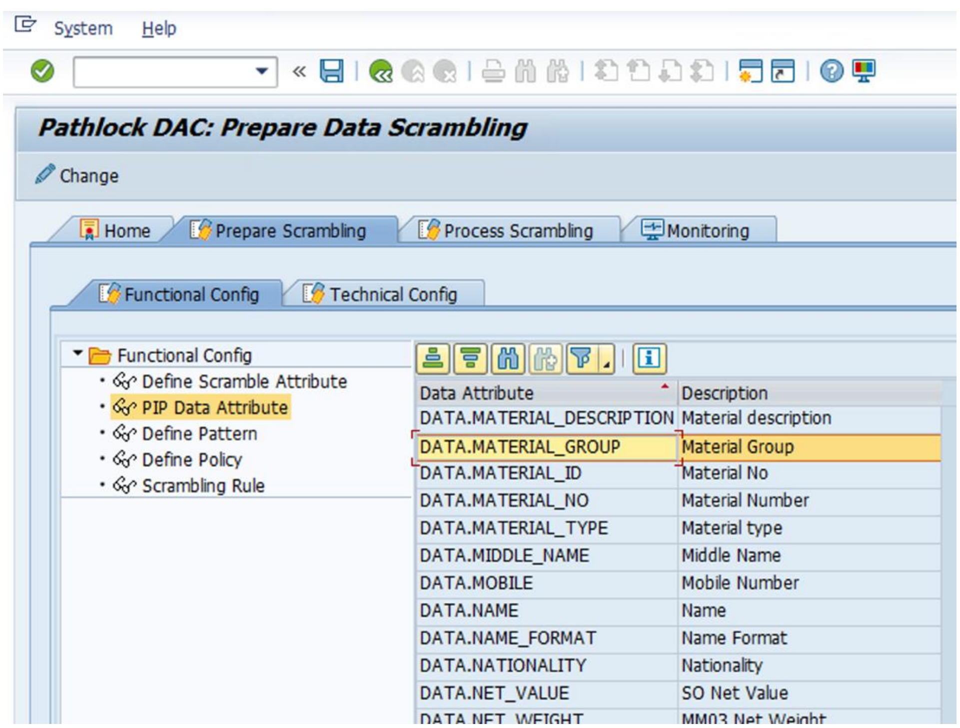
1. Define a Scramble Attribute **`SCRAMBLE.MATERIAL_DESCRIPTION`** and specify its description as **Material Description**.



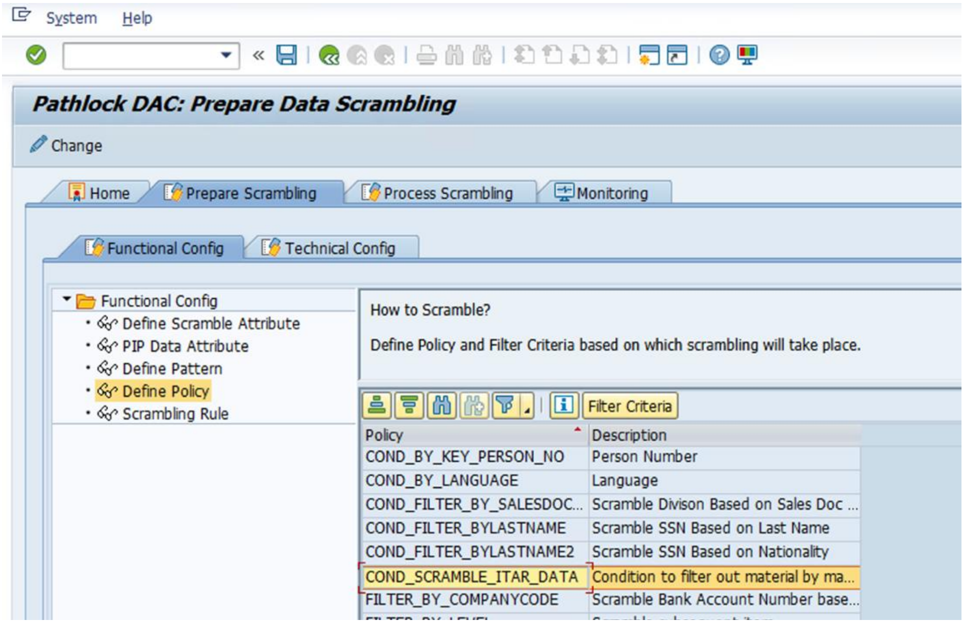
2. Define a Key Attribute **`KEY.MATERIAL`** and specify its description as **Material ID** for Scramble Attribute **`SCRAMBLE.MATERIAL_DESCRIPTION.`**



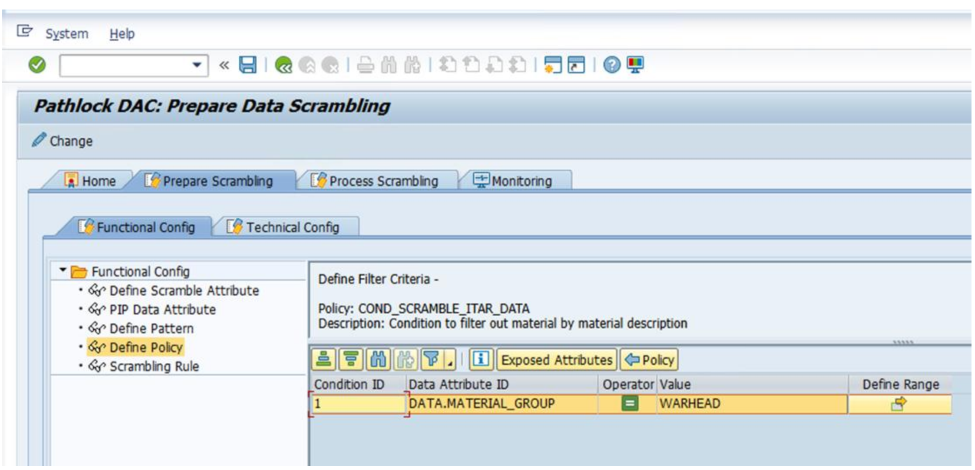
3. Define a **Data Attribute** with the Attribute ID **`DATA.MATERIAL_GROUP`** and specify its description as **Material Group**.



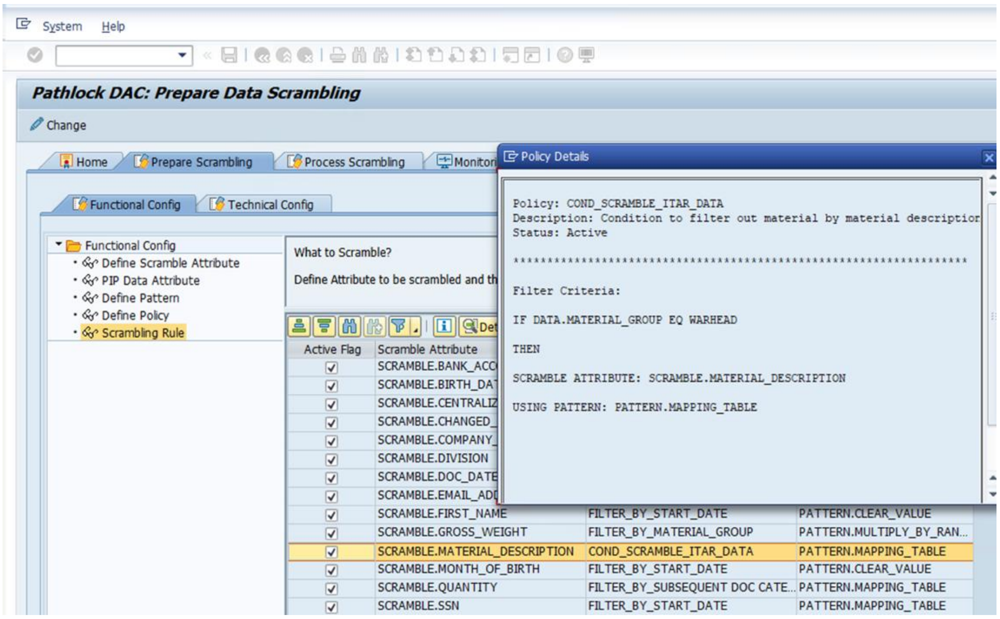
4. Create a **Policy** named **`COND_SCRAMBLE_ITAR_DATA`** and specify its description.



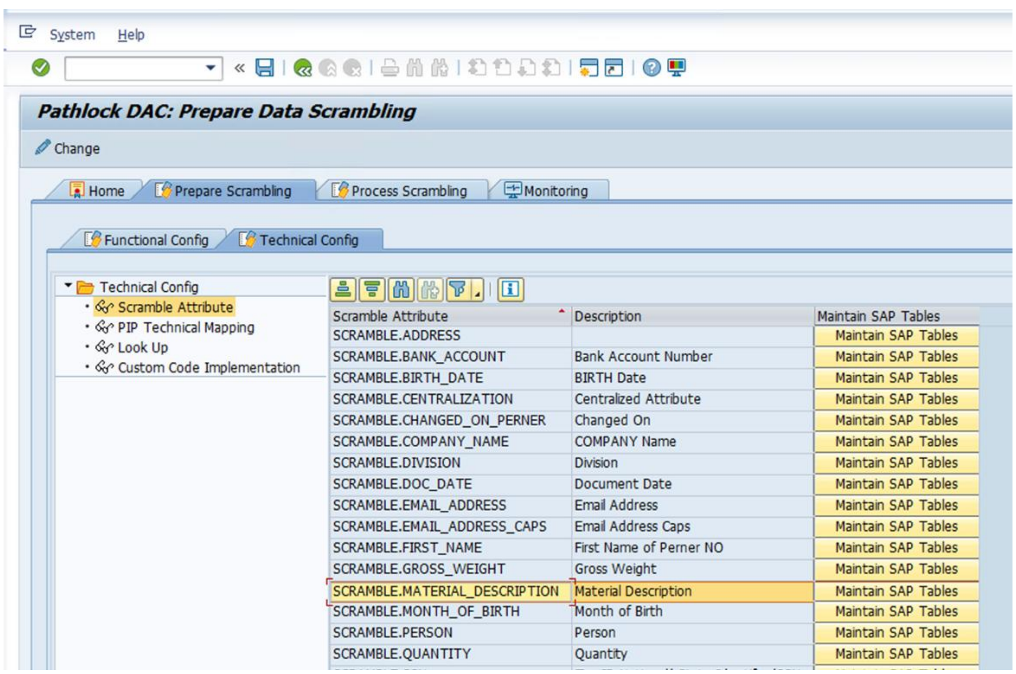
5. Define the Filter Criteria using the Data Attribute **`DATA.MATERIAL_GROUP`** within the Policy **`COND_SCRAMBLE_ITAR_DATA`.**

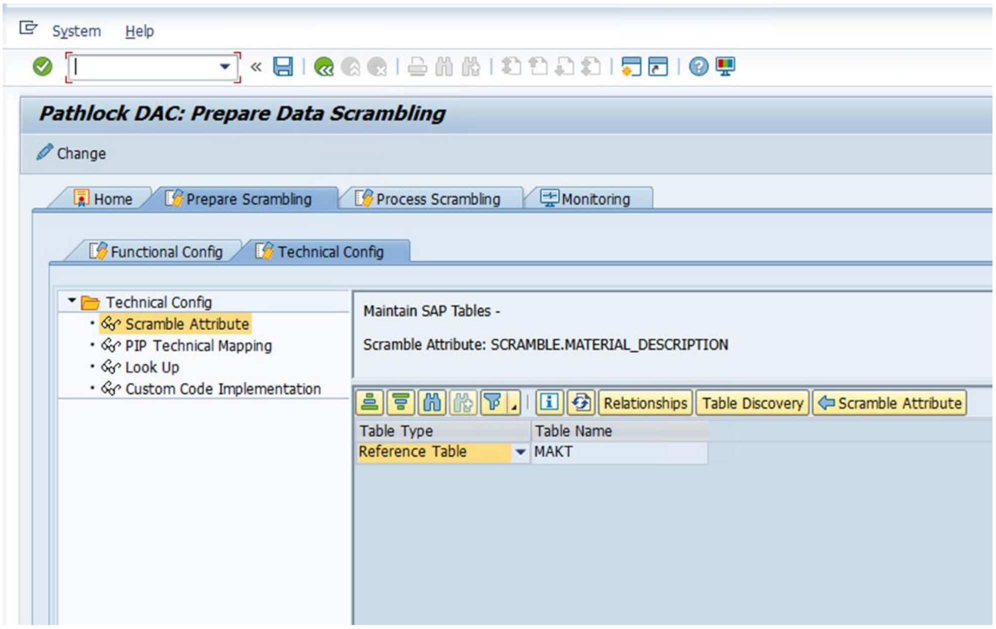


6. Configure the Scrambling Rule for the Scramble Attribute **`SCRAMBLE.MATERIAL_DESCRIPTION,`** linking it to Policy **`COND_SCRAMBLE_ITAR_DATA`** and the pattern **`PATTERN_MAPPING_TABLE.`**

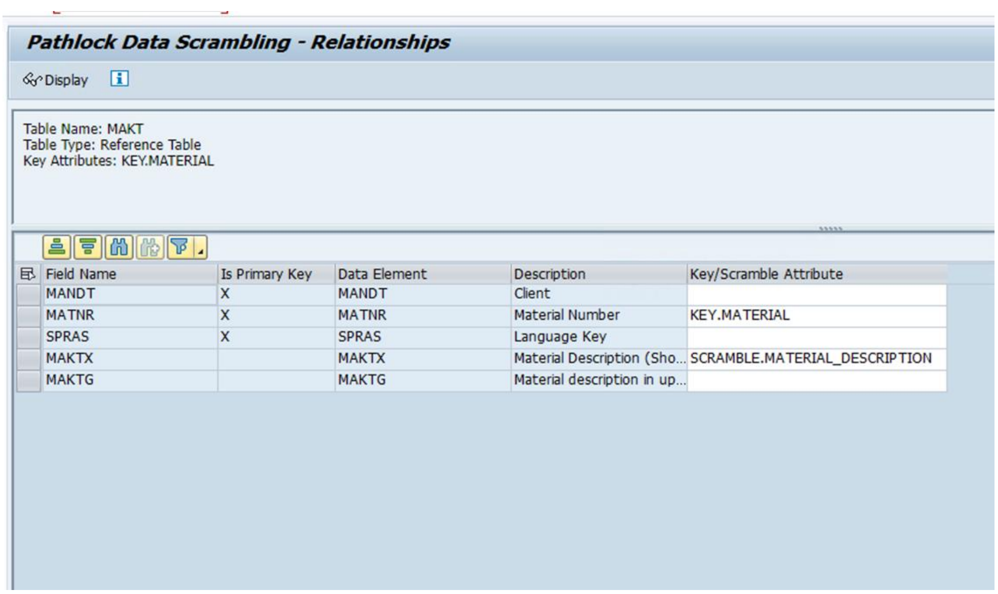


7. Under the **Technical Configuration** tab, maintain SAP tables for the Scramble Attribute **`SCRAMBLE.MATERIAL_DESCRIPTION.`**

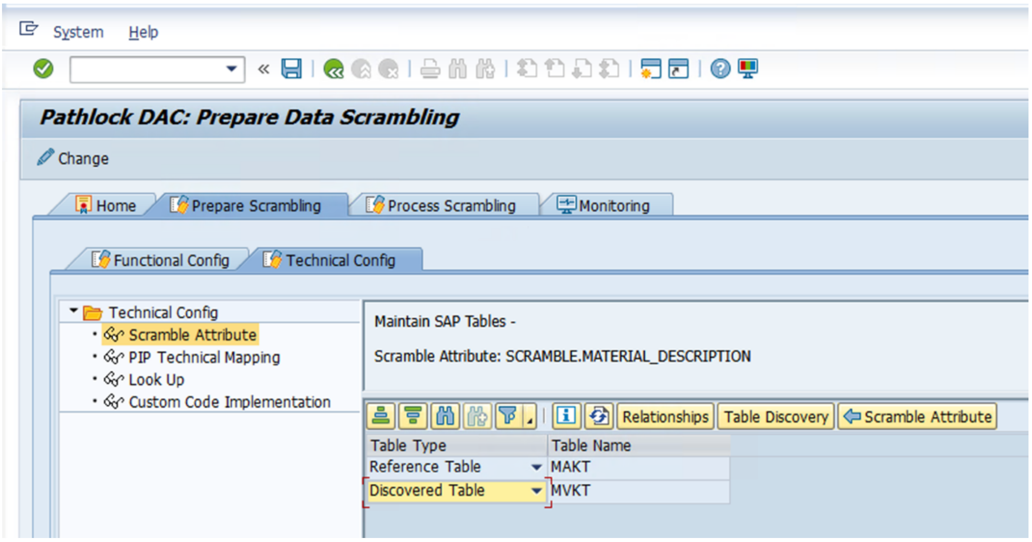




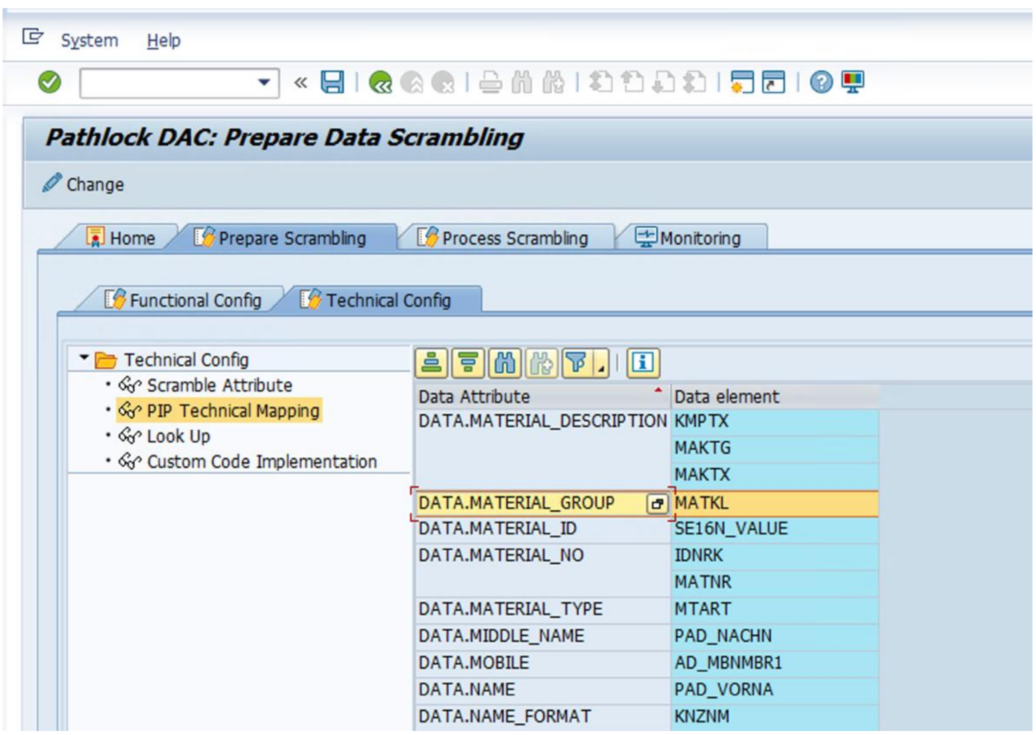
8. Select the reference table and establish relationships for the Scramble Attribute **`SCRAMBLE.MATERIAL_DESCRIPTION`** for field **`MAKTX`** and Key Attribute **`KEY.MATERIAL`** for field **`MATNR.`**



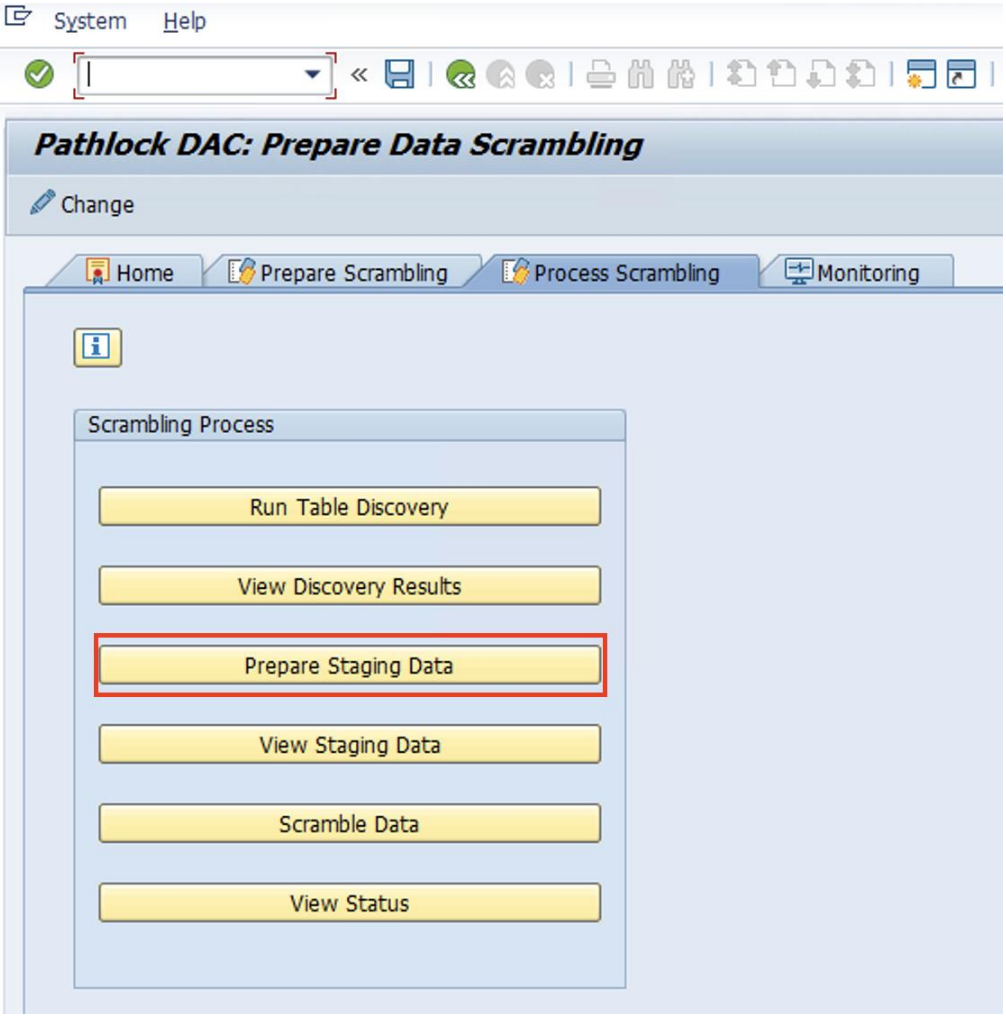
9. Click **Table Discovery** to get the discovered tables.



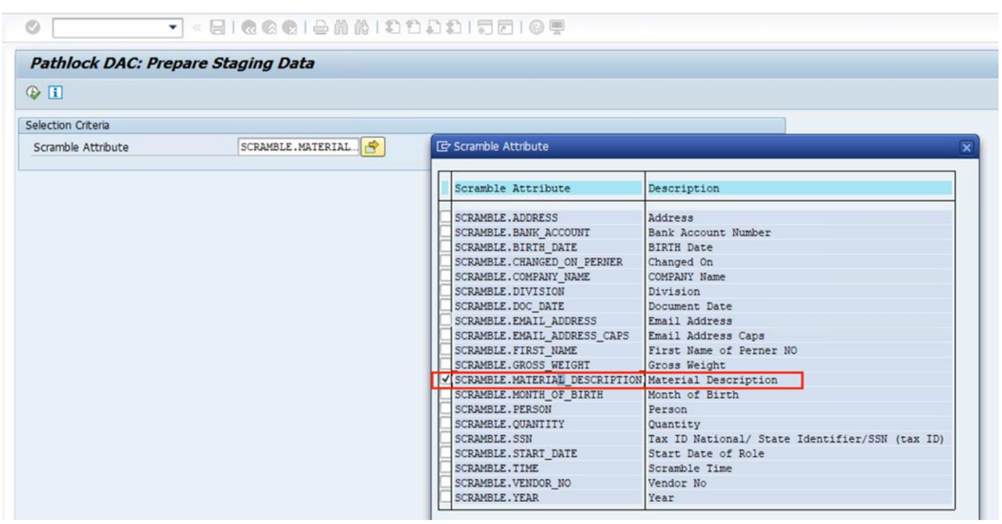
10. Under the **Technical Configuration** tab, define the Attribute **`DATA.MATERIAL_GROUP`** and map it to the Data Element **`MAKTL.`**



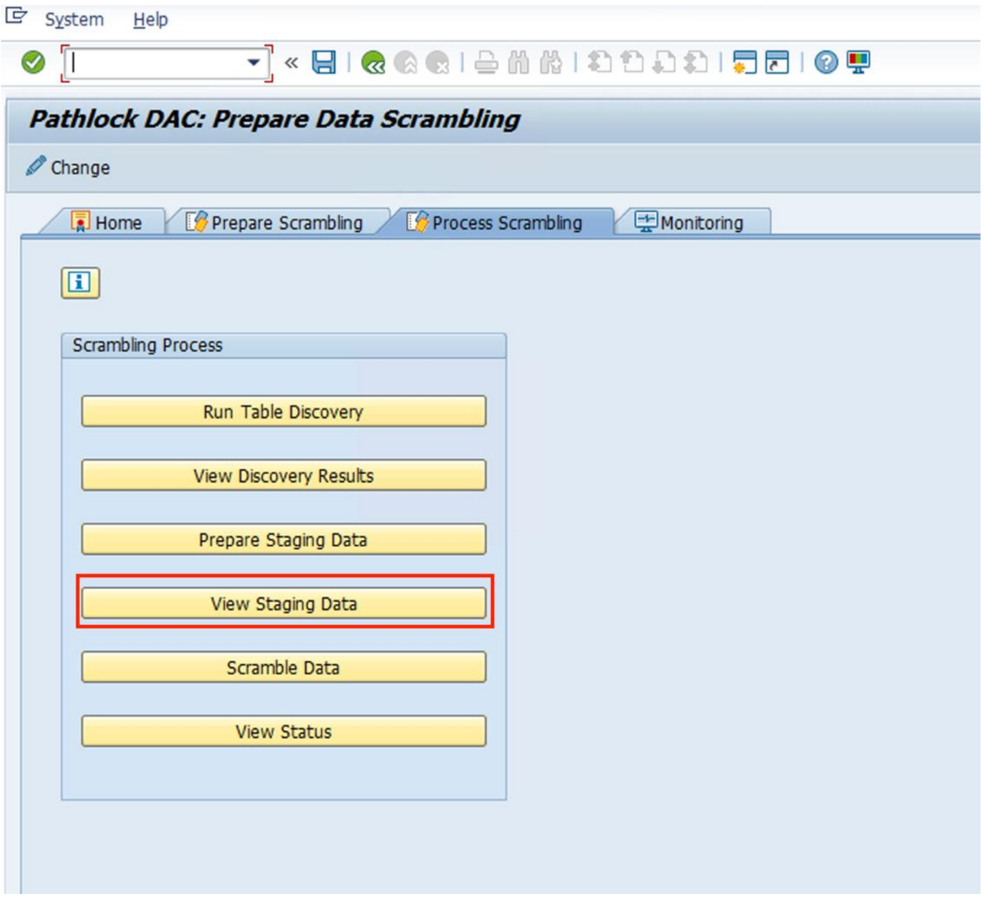
11. Navigate to the **Process Data Scrambling** tab to proceed with staging and scrambling the material attributes. Click **Prepare Staging Data** to stage the attributes,



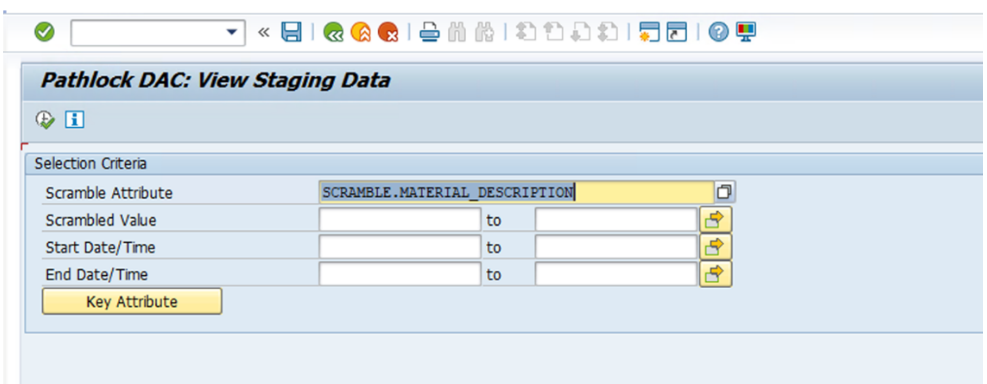
(a) Select the Scramble Attribute **`SCRAMBLE.MATERIAL_DESCRIPTION`** and execute.



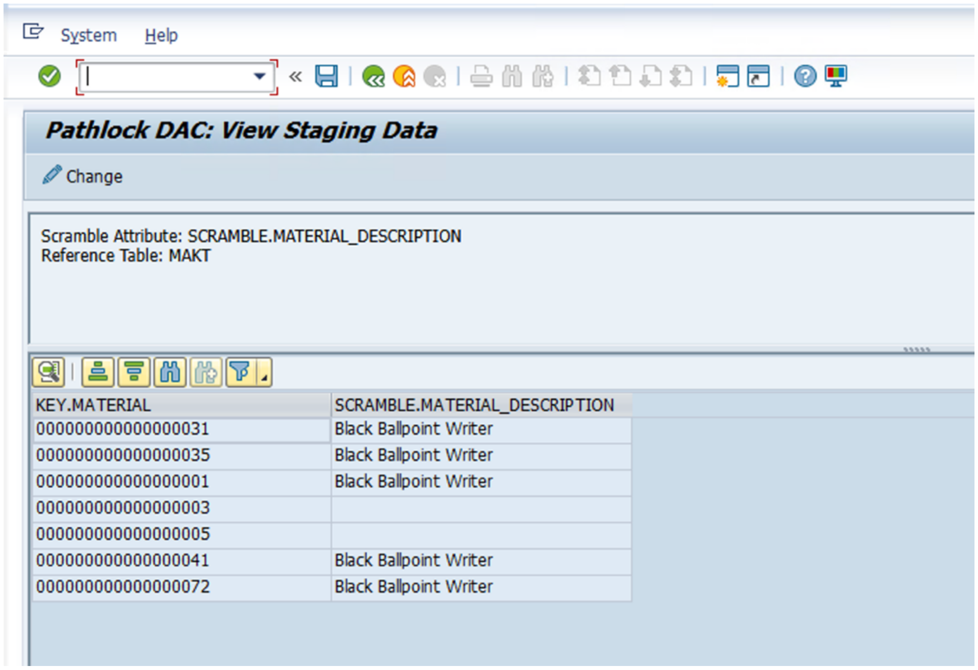
12. Click **View Staging Data** to review the staged data.



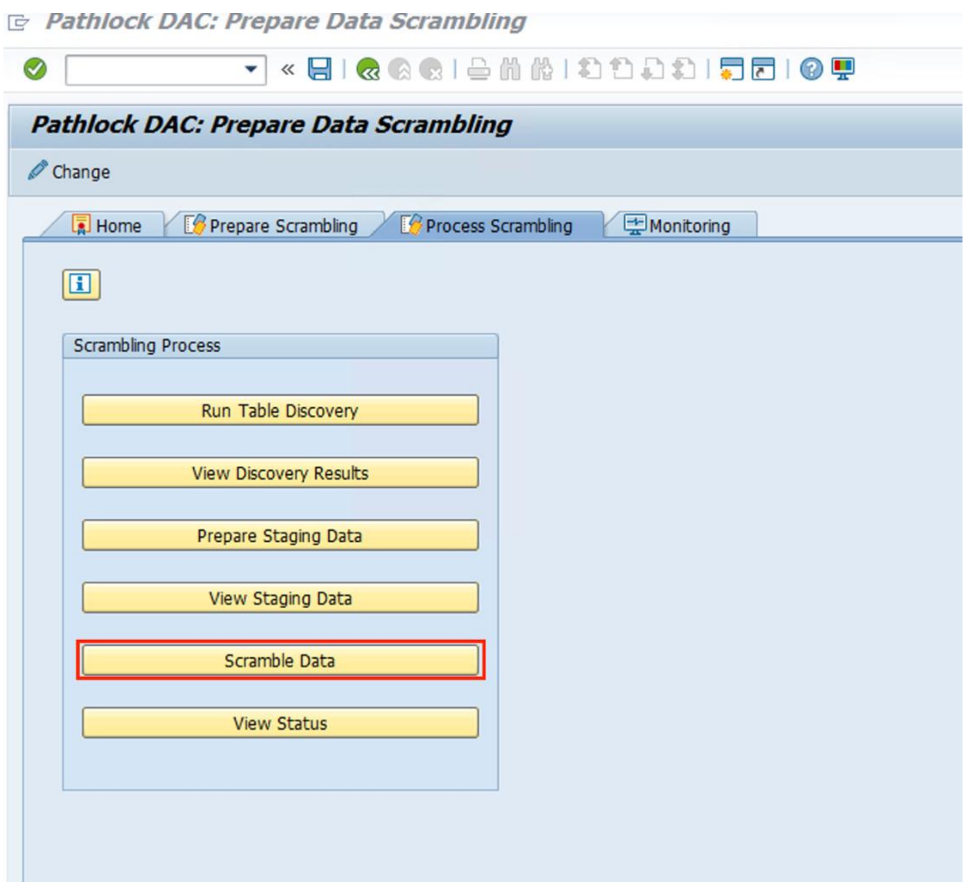
(a) Enter the Scramble Attribute `SCRAMBLE.MATERIAL_DESCRIPTION` and click **Execute** button to display the staging for the material description.



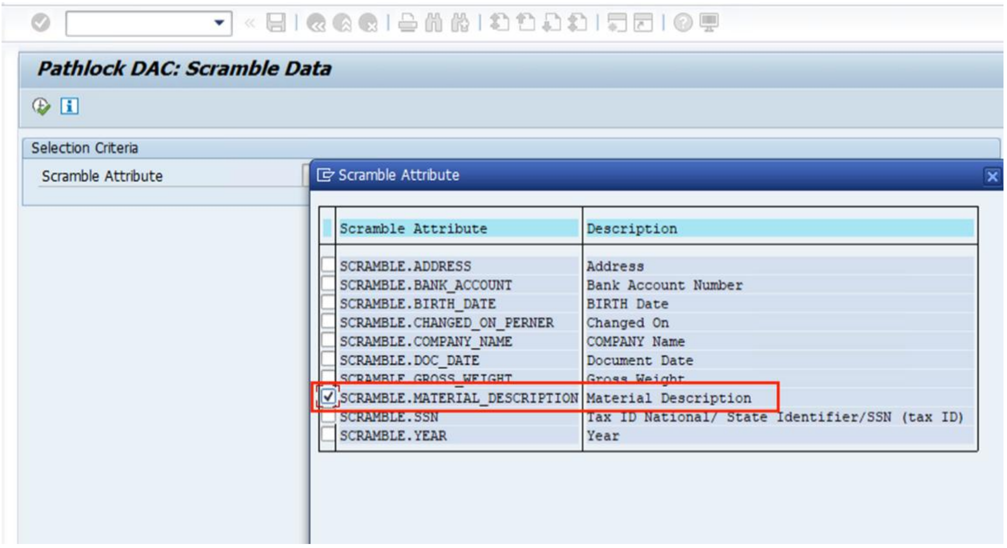
(b) The **Staging Area** allows reviewing the **Material Description** (Scramble Attribute) before scrambling it for each **Material Number** (Key Attribute).



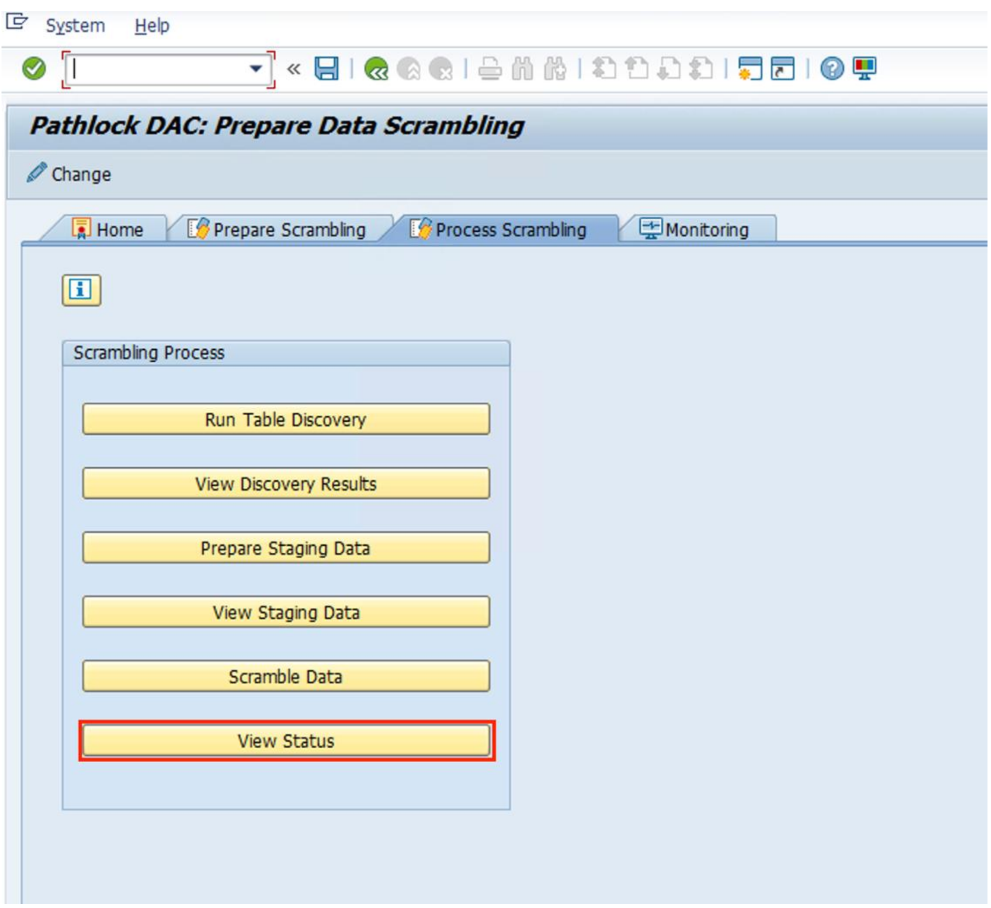
13. Click **Scramble Data** to scramble the material attributes.

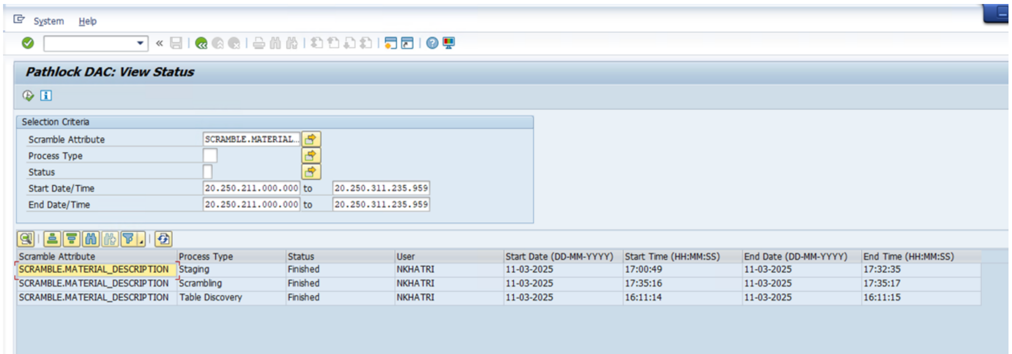


(a) Select the Scramble Attribute **`SCRAMBLE.MATERIAL_DESCRIPTION`** and click **Execute** button to apply scrambling to the material description.



14. To view the status of Table Discovery, Staging, or Scrambling Data, click **View Status**.

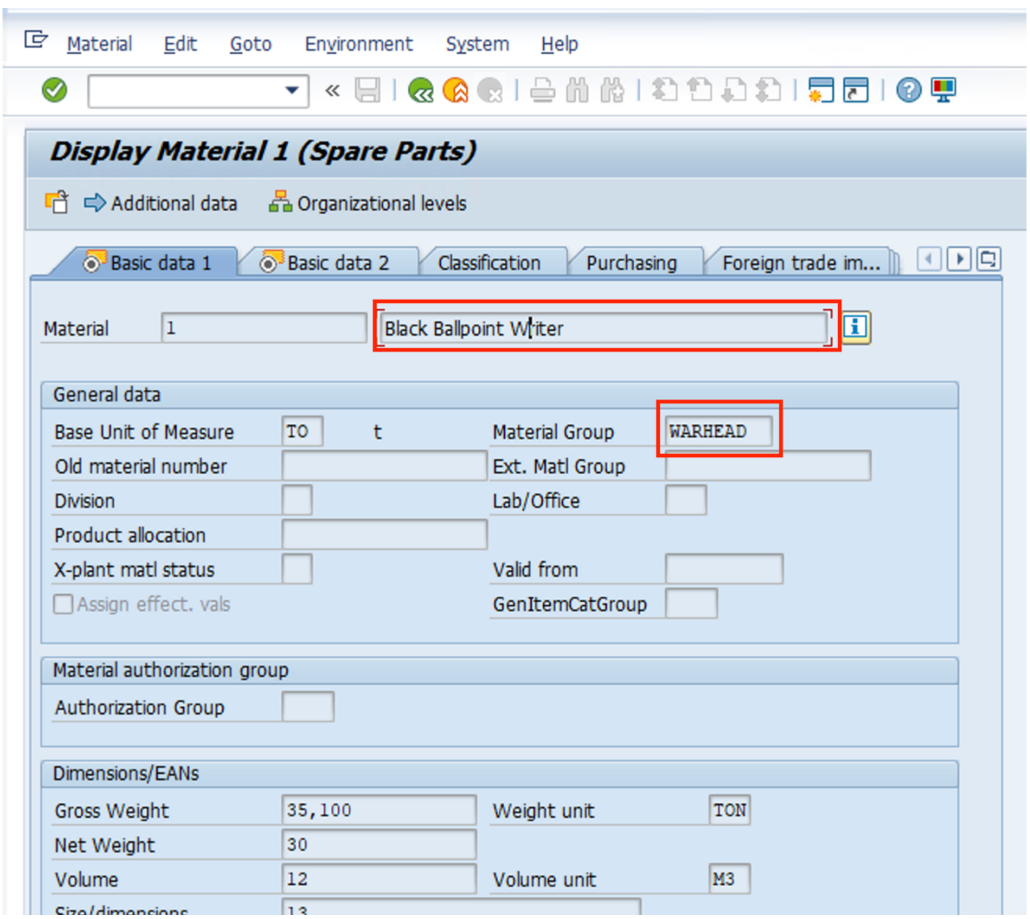




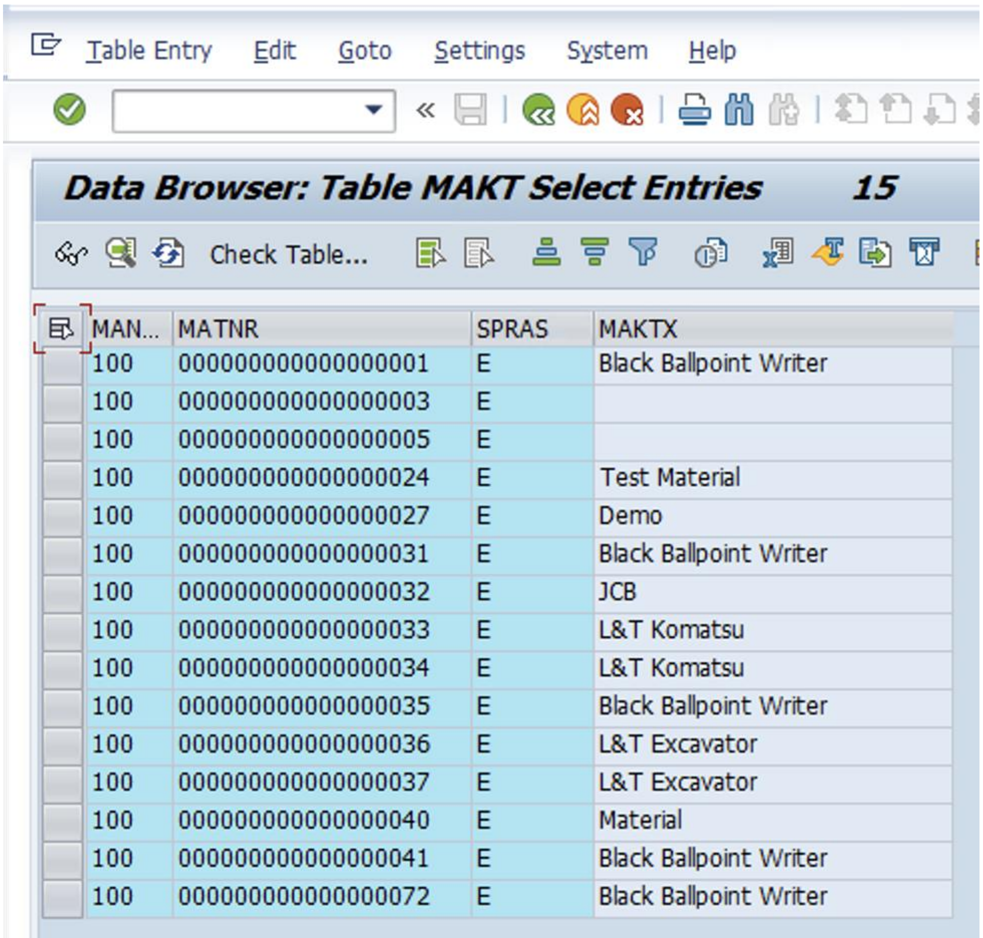
Expected Results:

After implementation, the following outcomes demonstrate the expected results of Data Scrambling. The scrambling will apply across all output formats, including GUI Transactions (MM03), SE16 table outputs, downloaded data in ALVs, search help, and any instance where the Data Element **MAKTL** is used:

- **Scrambling in Transaction ‘MM03’:** The **Material Description** field will appear scrambled in the MM03 transaction.



- **Scrambling in SE16 ALV Grid:** The **Material Description** field will be scrambled in the SE16 data browser.



MAN...	MATNR	SPRAS	MAKTX
100	000000000000000001	E	Black Ballpoint Writer
100	000000000000000003	E	
100	000000000000000005	E	
100	000000000000000024	E	Test Material
100	000000000000000027	E	Demo
100	000000000000000031	E	Black Ballpoint Writer
100	000000000000000032	E	JCB
100	000000000000000033	E	L&T Komatsu
100	000000000000000034	E	L&T Komatsu
100	000000000000000035	E	Black Ballpoint Writer
100	000000000000000036	E	L&T Excavator
100	000000000000000037	E	L&T Excavator
100	000000000000000040	E	Material
100	000000000000000041	E	Black Ballpoint Writer
100	000000000000000072	E	Black Ballpoint Writer